Borrowing Strength Without Borrowing Data: Online Supplement

Simulation study, real hospital networks and the design of GHB

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This supplement accompanies the paper *Borrowing Strength Without Borrowing Data: Private, Serverless and Tuning-Free Collaboration with Graph Hierarchical Borrowing* and follows its order. Section S1 gives the full design and results of the eighteen simulations of Section 6. Section S2 documents the TCGA study of Section 7.2: data, protocol, the configurations fixed for every literature method, the design results, the oracle arm, tuning GHB, and the design choices. Sections S3 and S4 document the FLamby benchmark and the image study, Section S5 every network's results for the small networks of Section 7.1, and Section S6 every hospital's results on UCI Heart Disease. Section S8 records how GHB-lite was designed and the extensions that were tried and not adopted. Every number is generated by the released scripts and read into the text as LaTeX macros.

S1 SIMULATION STUDY

Section 6 of the paper summarizes these simulations; this section gives the full design and every figure. The simulations test GHB-lite (Sections 4 and 5 of the paper) under its own model and under a logistic analogue. They check internal consistency and expose the cost of estimating the unknown quantities that Propositions 4.1 to 4.3 take as known; they are not evidence about real hospital networks. Unless stated otherwise, the network has 24 hospitals whose sizes are drawn once from a log-normal distribution with median 100 and clipped (16 to 479 patients), $\sigma^2 = 1$, $\delta = 10^{-5}$, clipping bound $C = 1$, and the privatized releases are averaged by 60 rounds of Metropolis-Hastings gossip on the ring with two-hop links of Figure 2. *Plug-in GHB-lite* estimates everything it needs from the privatized releases and public quantities only: τ^2 by DerSimonian-Laird [DerSimonian and Laird 1986] (Remark 4.4), the leave-one-out population estimate by inverse-variance weighting, and its error M from those weights. Risk is reported relative to local estimation, R_k/v_k with $v_k = \sigma^2/n_k$; values below 1 mean that a hospital gains. All code is in `simulations/` of the companion survey's repository, and `run_all.py` regenerates every figure and number in about twenty minutes.

Simulation 1: the closed forms. Figure S1 and Tables S1 and S2 repeat the setting of Figure 1 with 40,000 replicates per point. Across 101 comparisons of a simulated risk or M with its closed form, 98% lie within two Monte Carlo standard errors, and the largest relative deviation is 1.6%. Propositions 4.1 and 4.2 and the code that draws Figure 1 agree.

Simulation 2: heterogeneity. Figure S2 and Table S3 vary τ^2 from 10^{-4} to 1. Plug-in GHB-lite is never worse than local estimation by more than Monte Carlo error: its worst hospital reaches a relative risk of 1.02 at most. A single global model (the sample-size weighted mean of all site estimates, computed without privacy) is the best choice when τ^2 is tiny, but its mean relative risk reaches 134 and its worst hospital 382 at $\tau^2 = 1$. Estimating τ^2 costs most when heterogeneity is smallest: at $\tau^2 = 10^{-4}$ and $\epsilon = 1$ the plug-in reaches 0.12 against 0.07 with the true τ^2. Gossip with 60 rounds is indistinguishable from exact averaging.

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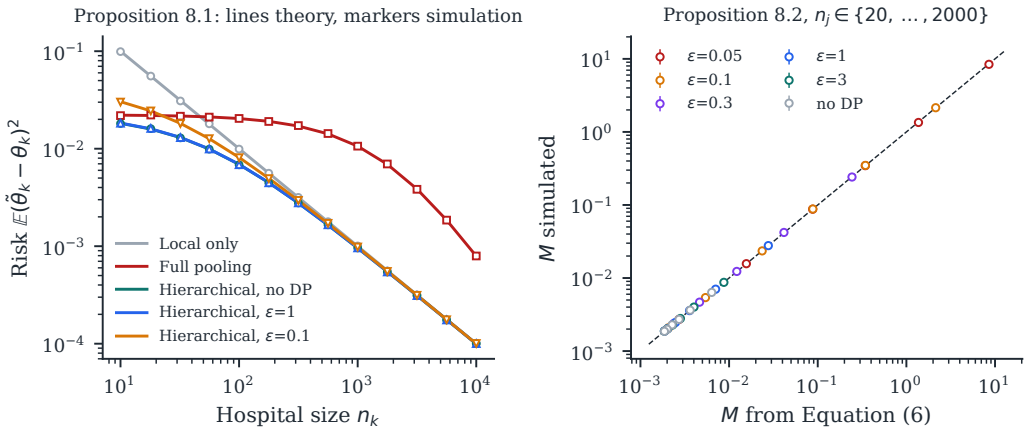


Fig. S1. Simulation 1. Left: risk against hospital size; lines are the closed forms of Figure 1, markers are simulated (±2 standard errors). Right: simulated M of the privatized leave-one-out mean against Equation (4), for 11 other hospitals of 20 to 2,000 patients and six privacy budgets.

Table S1. Simulation 1: exact values behind Figure S1. Top: risk R_k of a hospital of n_k patients ($\sigma^2 = 1$), simulated, with the closed form in grey; Global: full pooling without privacy; Hier.: hierarchical borrowing with known τ^2. Bottom: simulated M of the privatized leave-one-out mean, when the 23 other hospitals all have n_j patients, with Equation (4) in grey.

n_k	Local	Global	Hier., no DP	Hier., $\epsilon = 1$	Hier., $\epsilon = 0.1$
10	0.0991 (0.1000)	0.0218 (0.0221)	**0.0181** (0.0182)	0.0183 (0.0184)	0.0301 (0.0304)
18	0.0556 (0.0556)	0.0222 (0.0219)	**0.0161** (0.0159)	**0.0161** (0.0160)	0.0244 (0.0244)
32	0.0309 (0.0312)	0.0215 (0.0216)	**0.0129** (0.0130)	0.0130 (0.0131)	0.0181 (0.0182)
56	0.0180 (0.0179)	0.0211 (0.0212)	**0.0099** (0.0099)	**0.0099** (0.0100)	0.0127 (0.0127)
100	0.0099 (0.0100)	0.0204 (0.0204)	**0.0068** (0.0069)	**0.0068** (0.0069)	0.0081 (0.0081)
178	0.0056 (0.0056)	0.0190 (0.0191)	**0.0045** (0.0045)	**0.0045** (0.0045)	0.0049 (0.0050)
316	0.0032 (0.0032)	0.0173 (0.0171)	**0.0028** (0.0028)	**0.0028** (0.0028)	0.0030 (0.0030)
562	0.0018 (0.0018)	0.0143 (0.0142)	**0.0017** (0.0016)	**0.0017** (0.0016)	**0.0017** (0.0017)
1,000	**0.0010** (0.0010)	0.0106 (0.0106)	**0.0010** (0.0010)	**0.0010** (0.0010)	**0.0010** (0.0010)
1,778	0.0006 (0.0006)	0.0070 (0.0069)	**0.0005** (0.0005)	**0.0005** (0.0005)	0.0006 (0.0006)
3,162	**0.0003** (0.0003)	0.0038 (0.0039)	**0.0003** (0.0003)	**0.0003** (0.0003)	**0.0003** (0.0003)
5,623	**0.0002** (0.0002)	0.0019 (0.0019)	**0.0002** (0.0002)	**0.0002** (0.0002)	**0.0002** (0.0002)
10,000	**0.0001** (0.0001)	0.0008 (0.0008)	**0.0001** (0.0001)	**0.0001** (0.0001)	**0.0001** (0.0001)

Table S2. Simulation 1, continued: M against the size n_j of the other hospitals and the privacy budget; simulated, with Equation (4) in grey.

n_j	$\epsilon = 0.05$	$\epsilon = 0.1$	$\epsilon = 0.3$	$\epsilon = 1$	$\epsilon = 3$	no DP
20	8.42 (8.54)	2.14 (2.14)	0.241 (0.243)	0.0278 (0.0277)	0.00872 (0.00873)	0.00636 (0.00636)
50	1.35 (1.37)	0.348 (0.345)	0.042 (0.0416)	0.00703 (0.00705)	0.004 (0.00402)	0.00363 (0.00364)
100	0.345 (0.344)	0.088 (0.0881)	0.0123 (0.0122)	0.00358 (0.00358)	0.00279 (0.00282)	0.00272 (0.00273)
200	0.0875 (0.0876)	0.0235 (0.0236)	0.00467 (0.00464)	0.00247 (0.00249)	0.00227 (0.0023)	0.00228 (0.00227)
500	0.0157 (0.0157)	0.00539 (0.00541)	0.00239 (0.00238)	0.00204 (0.00203)	0.002 (0.002)	0.00199 (0.002)
2,000	0.0027 (0.00272)	0.00209 (0.00208)	0.00189 (0.00189)	0.00188 (0.00187)	0.00188 (0.00186)	0.00186 (0.00186)

Simulation 3: who gains, and how many hospitals are needed. At $\tau^2 = 0.02$ and $\epsilon = 1$ (Figure S3, left), every hospital gains about the same number of virtual patients, $B \approx 46$, as Equation (3) predicts. The hospitals with 60 patients or fewer multiply their effective sample size by 2.28 on average,

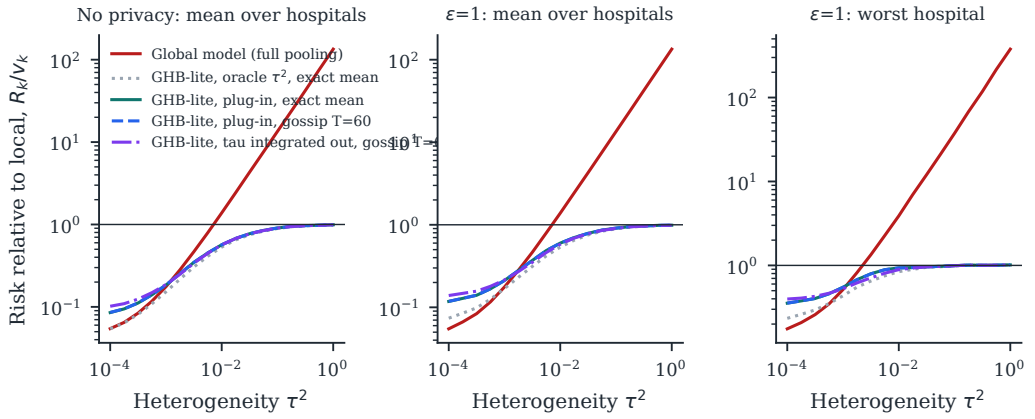


Fig. S2. Simulation 2. Risk relative to local estimation as a function of heterogeneity, mean over hospitals without privacy and with $\epsilon = 1$, and for the worst hospital with $\epsilon = 1$.

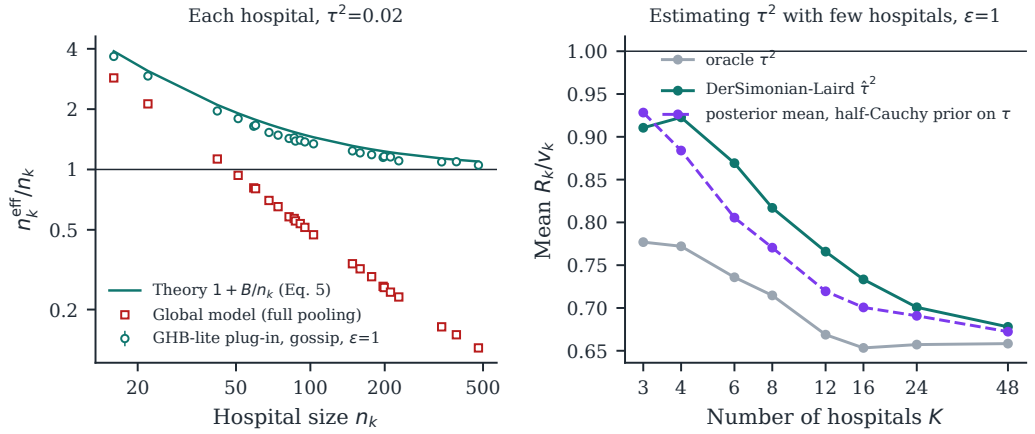


Fig. S3. Simulation 3. Left: effective sample size relative to own size for each hospital, plug-in GHB-lite and a global model; the line is $1 + B/n_k$. Right: mean relative risk against the number of hospitals for three ways of obtaining τ^2.

and those with 300 or more by 1.08; the smallest gain of any hospital is 1.05. The global model multiplies the effective sample size of small hospitals by 1.44 but cuts that of large hospitals to 0.15 of their own. With few hospitals, τ^2 is poorly estimated (Figure S3, right, and Table S4). With three hospitals the mean relative risk is 0.91 with DerSimonian-Laird [DerSimonian and Laird 1986] against 0.78 with the true τ^2, and the gap closes as the network grows (0.68 against 0.66 with 48 hospitals). The posterior mean of τ^2 under a half-Cauchy(0.5) prior on τ, the weakly informative choice that Remark 4.4 recommends, does better than DerSimonian-Laird [DerSimonian and Laird 1986] from four hospitals upward but not with three.

Simulation 4: gossip rounds. Figure S4 and Table S5 show the mean relative risk of plug-in GHB-lite after T rounds of gossip on the four topologies of Figure 2. The risk comes within 2% of its

Table S3. Simulation 2: exact values behind Figure S2. Relative risk R_k/v_k, mean over hospitals and worst hospital, at $\epsilon = 1$ (top) and without privacy (bottom). Global: full pooling, without privacy in both blocks; Oracle: GHB-lite with the true τ^2 and exact averaging; Plug-in: DerSimonian-Laird [DerSimonian and Laird 1986], with exact averaging or 60 gossip rounds; FB: τ integrated out (Remark 4.4), 60 gossip rounds.

	Global		Oracle		Plug-in, exact		Plug-in, gossip		FB, gossip	
τ^2	mean	worst	mean	worst	mean	worst	mean	worst	mean	worst
0.0001	**0.055**	**0.18**	0.074	0.23	0.118	0.35	0.118	0.36	0.139	0.40
0.00018	**0.066**	**0.21**	0.085	0.26	0.128	0.38	0.129	0.38	0.148	0.41
0.00032	**0.084**	**0.26**	0.099	0.30	0.140	0.40	0.140	0.40	0.157	0.43
0.00056	**0.117**	**0.35**	0.126	**0.35**	0.168	0.46	0.168	0.46	0.180	0.47
0.001	0.176	0.52	**0.169**	**0.43**	0.212	0.55	0.212	0.55	0.215	0.53
0.0018	0.282	0.82	**0.231**	**0.55**	0.279	0.68	0.279	0.68	0.271	0.62
0.0032	0.465	1.34	**0.313**	**0.65**	0.370	0.80	0.370	0.79	0.351	0.71
0.0056	0.799	2.28	**0.420**	**0.75**	0.488	0.89	0.487	0.89	0.462	0.81
0.01	1.383	3.89	**0.536**	**0.84**	0.600	0.93	0.599	0.93	0.581	0.89
0.018	2.435	6.97	**0.651**	**0.90**	0.700	0.94	0.700	0.94	0.689	0.93
0.032	4.284	12.08	**0.754**	**0.95**	0.788	0.96	0.788	0.96	0.780	0.96
0.056	7.593	21.17	**0.837**	**0.96**	0.863	0.97	0.863	0.97	0.854	0.97
0.1	13.470	37.47	**0.894**	**0.99**	0.911	**0.99**	0.911	**0.99**	0.904	**0.99**
0.18	23.878	67.75	**0.938**	**1.00**	0.950	1.01	0.950	1.01	0.944	1.01
0.32	42.519	118.15	**0.960**	**1.00**	0.968	**1.00**	0.968	**1.00**	0.964	**1.00**
0.56	75.568	216.32	**0.979**	**1.00**	0.984	1.01	0.984	1.01	0.981	1.01
1	134.283	377.26	**0.985**	**1.01**	0.988	**1.01**	0.988	**1.01**	0.987	**1.01**
0.0001	**0.054**	**0.17**	**0.054**	**0.17**	0.085	0.27	0.086	0.27	0.102	0.31
0.00018	0.065	**0.20**	**0.064**	**0.20**	0.094	0.29	0.095	0.29	0.109	0.32
0.00032	0.084	0.26	**0.081**	**0.24**	0.111	0.33	0.112	0.33	0.124	0.35
0.00056	0.118	0.36	**0.109**	**0.31**	0.140	0.40	0.141	0.40	0.149	0.41
0.001	0.176	0.51	**0.152**	**0.40**	0.184	0.49	0.184	0.49	0.186	0.47
0.0018	0.281	0.82	**0.217**	**0.51**	0.253	0.62	0.253	0.62	0.248	0.57
0.0032	0.467	1.35	**0.305**	**0.64**	0.347	0.75	0.346	0.75	0.338	0.70
0.0056	0.794	2.27	**0.411**	**0.75**	0.452	0.84	0.452	0.83	0.447	0.80
0.01	1.381	3.86	**0.532**	**0.83**	0.567	0.88	0.567	0.88	0.567	0.87
0.018	2.430	6.86	**0.649**	**0.89**	0.677	0.91	0.677	0.91	0.676	0.91
0.032	4.312	12.12	**0.751**	**0.95**	0.773	0.96	0.773	0.96	0.770	0.96
0.056	7.609	21.66	**0.830**	**0.96**	0.848	0.97	0.847	0.97	0.844	0.97
0.1	13.479	39.01	**0.894**	**0.98**	0.907	**0.98**	0.907	**0.98**	0.903	**0.98**
0.18	23.800	66.42	**0.939**	**0.99**	0.948	1.00	0.948	1.00	0.944	**0.99**
0.32	42.257	118.05	**0.963**	**1.01**	0.969	**1.01**	0.969	**1.01**	0.966	**1.01**
0.56	75.882	215.48	**0.976**	**1.00**	0.979	**1.00**	0.979	**1.00**	0.978	**1.00**
1	134.283	382.26	**0.990**	**1.02**	0.992	**1.02**	0.992	**1.02**	0.991	**1.02**

Table S4. Simulation 3: exact values behind Figure S3, right. Mean relative risk against the number of hospitals K, for the true τ^2, DerSimonian-Laird [DerSimonian and Laird 1986], and the posterior mean under a half-Cauchy(0.5) prior on τ.

K	True τ^2	DerSimonian-Laird [DerSimonian and Laird 1986]	Posterior mean
3	**0.777**	0.911	0.928
4	**0.772**	0.923	0.884
6	**0.736**	0.869	0.806
8	**0.715**	0.817	0.770
12	**0.669**	0.766	0.719
16	**0.653**	0.733	0.701
24	**0.657**	0.701	0.691
48	**0.658**	0.678	0.672

exact-average value after 35 rounds on the ring, 22 on the regional clusters, 11 on the sparse random graph and 14 on the ring with two-hop links, far fewer than the rounds that the worst-case bound of Proposition 4.3 requires for a consensus error of 1% (203, 127, 73 and 69). The bound is conservative

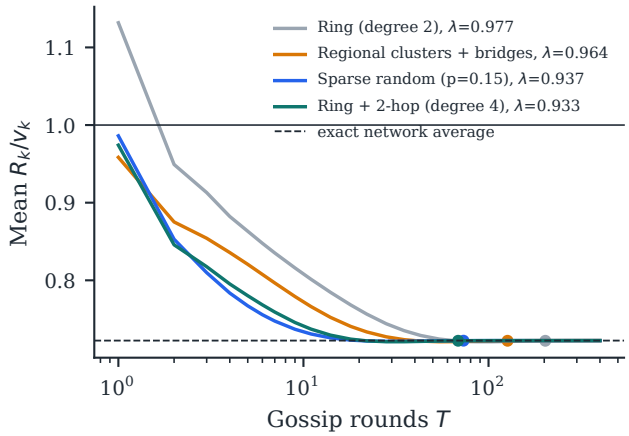


Fig. S4. Simulation 4. Mean relative risk of plug-in GHB-lite against the number of gossip rounds ($\tau^2 = 0.02$, $\epsilon = 1$). Dots mark $\log(100)/(1-\lambda)$; the dashed line is the exact network average.

for the risk: it controls the consensus error of the releases, while the risk depends on that error only through a population estimate that the shrinkage weight already discounts. After a single round on the ring, GHB-lite is worse than local estimation (relative risk 1.13), so a hospital should not use the borrowed estimate before the gossip has mixed.

Simulation 5: joint release by small hospitals. Section 4.2 proposes that small hospitals release jointly through secure aggregation, so that the privacy noise is calibrated to the group. Figure S5 and Table S6 show that the remedy matters only when every hospital is small. In a network of 16 hospitals of 20 patients and 8 of 200, the large hospitals dominate the inverse-variance weighted population estimate, and groups of eight raise the borrowed patients only from 19.3 to 21.0 at $\epsilon = 0.1$. In a network of 24 hospitals of 20 patients, the same grouping raises B from 1.0 to 6.9 at $\epsilon = 0.1$ and from 30.6 to 41.4 at $\epsilon = 1$.

Simulation 6: structured heterogeneity. The 24 hospitals are split into four regions of six; θ_k is the sum of a regional effect with variance τ_r^2 and a hospital effect with variance $\tau_w^2 = 0.002$. The GMRF prior of Equation (5) links hospitals of the same region and is computed by Jacobi iterations on the privatized releases, followed by each hospital's local update with its own noise-free estimate. Its anchor term is switched off so that it uses the same information as the exchangeable prior, and its smoothing weight β was chosen on separate tuning replicates (the same value, $\beta \approx 46$, was selected at every τ_r^2); the regions are assumed known. Figure S6 and Table S7 show a crossover. When regions barely differ ($\tau_r^2 \lesssim 0.003$), the exchangeable prior is better (0.31 against 0.40 at the smallest τ_r^2), because the GMRF borrows only from five neighbors. When regional differences dominate, the exchangeable prior loses its gain (0.94) while the GMRF keeps it (0.40).

Simulation 7: logistic-regression heads. This simulation leaves the Gaussian model. Each hospital has its own logistic model on four standardized features and an intercept, $\theta_k = \mu + u_k$ with $u_k \sim \mathcal{N}(0, \tau^2 I)$, and features are clipped so that $\|x\| \le \sqrt{5}$. Here μ_0 is a published model (channel P3), equal to μ plus noise of standard deviation 0.3. Each hospital releases its mean score at μ_0, $g_k = \frac{1}{n_k}\sum_i x_i(y_i - \sigma(x_i^\top \mu_0))$, plus isotropic Gaussian noise. Its ℓ_2 sensitivity is $2\sqrt{5}/n_k$, and the noise is calibrated with zero-concentrated DP: $s_k = \Delta_k/\sqrt{2\rho}$ with $\rho + 2\sqrt{\rho \ln(1/\delta)} = \epsilon$. Under the

Table S5. Simulation 4: exact values behind Figure S4. Mean relative risk of plug-in GHB-lite after T gossip rounds on each topology; below, the second-largest eigenvalue modulus λ of the gossip matrix, the rounds after which the risk is within 2% of exact averaging, and the rounds that Proposition 4.3 requires for a consensus error of 1%.

T	Ring	Regional	Sparse random	Ring + 2-hop
1	1.132	**0.958**	0.986	0.974
2	0.949	0.875	0.853	**0.846**
3	0.913	0.854	**0.810**	0.818
4	0.882	0.836	**0.784**	0.795
5	0.863	0.821	**0.767**	0.780
6	0.848	0.808	**0.756**	0.768
7	0.835	0.797	**0.748**	0.759
9	0.816	0.779	**0.737**	0.746
11	0.801	0.766	**0.731**	0.737
14	0.785	0.752	**0.726**	0.729
18	0.768	0.741	**0.723**	0.724
22	0.756	0.733	**0.722**	**0.722**
28	0.744	0.727	**0.721**	**0.721**
35	0.735	0.724	**0.721**	**0.721**
43	0.729	0.722	0.722	**0.721**
54	0.724	**0.721**	0.722	0.722
68	0.722	**0.721**	0.722	0.722
85	**0.721**	**0.721**	0.722	0.722
106	**0.721**	0.722	0.722	0.722
132	**0.721**	0.722	0.722	0.722
165	**0.722**	**0.722**	**0.722**	**0.722**
206	**0.722**	**0.722**	**0.722**	**0.722**
257	**0.722**	**0.722**	**0.722**	**0.722**
320	**0.722**	**0.722**	**0.722**	**0.722**
400	**0.722**	**0.722**	**0.722**	**0.722**
λ	0.977	0.964	0.937	**0.933**
rounds to 2%	35	22	**11**	14
bound, $\gamma = 0.01$	203	127	73	**69**
exact averaging			0.722	

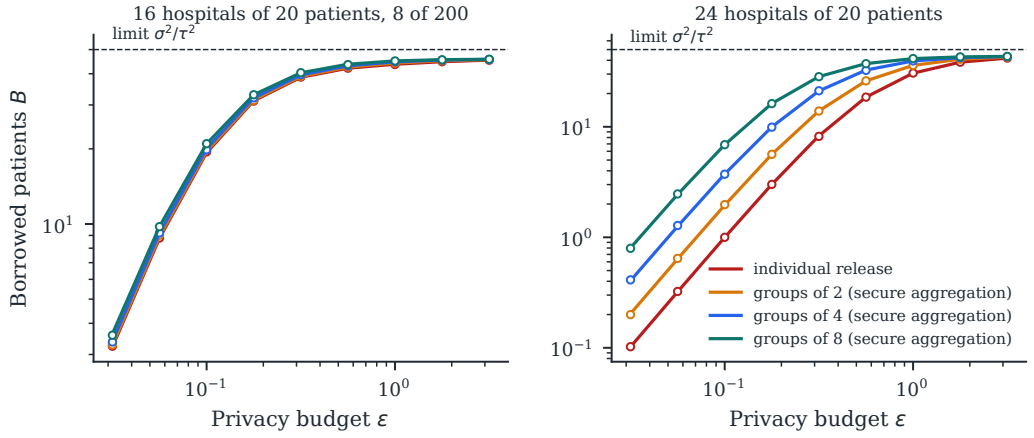


Fig. S5. Simulation 5. Borrowed patients B against the privacy budget for individual and joint release; lines are closed forms, markers simulation.

public feature distribution, $\mathbb{E}g_k = m(\theta_k) - m(\mu_0)$ with $m(\theta) = \mathbb{E}[x\,\sigma(x^\top\theta)]$, so the releases are mapped to the parameter scale by moment matching, linearized first at the pooled moment estimate

Table S6. Simulation 5: exact values behind Figure S5. Borrowed patients B for groups of g hospitals releasing jointly, simulated, with the closed form in parentheses. All small: 24 hospitals of 20 patients; mixed: 16 of 20 and 8 of 200 ($\tau^2 = 0.02$).

Network	ϵ	$g = 1$	$g = 2$	$g = 4$	$g = 8$
all small	0.0316	0.1 (0.1)	0.2 (0.2)	0.4 (0.4)	0.8 (0.8)
	0.0562	0.3 (0.3)	0.6 (0.6)	1.3 (1.3)	2.5 (2.4)
	0.1	1.0 (1.0)	2.0 (2.0)	3.7 (3.7)	6.9 (6.9)
	0.178	3.0 (3.0)	5.6 (5.6)	9.9 (10.0)	16.2 (16.2)
	0.316	8.2 (8.3)	13.9 (13.9)	21.2 (21.1)	28.4 (28.5)
	0.562	18.6 (18.6)	26.1 (26.1)	32.4 (32.6)	37.3 (37.3)
	1	30.7 (30.6)	35.9 (36.0)	39.5 (39.4)	41.5 (41.4)
	1.78	38.4 (38.4)	40.9 (40.9)	42.3 (42.2)	42.9 (42.9)
	3.16	41.9 (41.9)	42.6 (42.7)	43.1 (43.2)	43.3 (43.4)
mixed	0.0316	3.2 (3.2)	3.3 (3.3)	3.4 (3.4)	3.6 (3.6)
	0.0562	8.8 (8.8)	9.0 (8.9)	9.2 (9.2)	9.8 (9.8)
	0.1	19.5 (19.3)	19.8 (19.6)	19.8 (20.1)	21.0 (21.0)
	0.178	31.1 (31.2)	31.3 (31.5)	32.0 (32.0)	33.0 (33.0)
	0.316	38.8 (38.7)	39.1 (39.0)	39.6 (39.6)	40.4 (40.4)
	0.562	42.1 (42.1)	42.6 (42.5)	43.1 (43.0)	43.5 (43.7)
	1	43.6 (43.7)	44.1 (44.2)	44.7 (44.6)	45.0 (45.0)
	1.78	44.7 (44.7)	45.1 (45.1)	45.4 (45.4)	45.5 (45.6)
	3.16	45.3 (45.4)	45.6 (45.6)	45.6 (45.7)	45.7 (45.7)

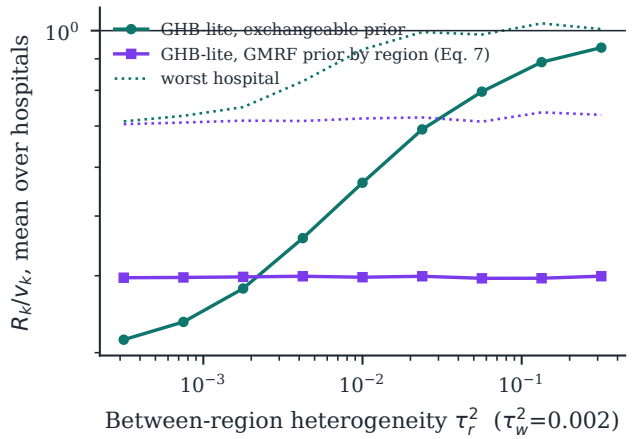


Fig. S6. Simulation 6. Mean relative risk (solid) and worst hospital (dotted) for the exchangeable and the regional GMRF prior, against the between-region variance ($\epsilon = 1$).

and then at each hospital's leave-one-out network estimate; their variances follow by the delta method. The anchor shapes that second point only for hospitals where it passes the conflict check against the anchor-free network estimate. GHB-lite then borrows coordinatewise. Each hospital summarizes the posterior of τ (half-Cauchy(0.5) prior) by 12 quantiles, combines each component's network estimate with μ_0 by precision weighting unless it dropped the anchor, multiplies by the weakly informative default $\mathcal{N}(0, 2.5^2)$ [Gelman et al. 2008], and averages its maximum a posteriori fits under the components, weighted by the Laplace approximation of the marginal likelihood of its own data (Remark 4.4). The code is `simulations/ghb_logit.py`. The baselines are a local model with prior $\mathcal{N}(\mu_0, I)$, which uses the anchor but no other hospital, and a global model fitted on the pooled data of all hospitals without privacy. Table S8 and Figure S7 report the excess log-loss over

Table S7. Simulation 6: exact values behind Figure S6. Mean and worst relative risk for the exchangeable prior and the regional GMRF prior against the between-region variance τ_r^2 ($\epsilon = 1$).

	Mean		Worst hospital	
τ_r^2	Exchangeable	GMRF	Exchangeable	GMRF
0.00032	**0.315**	0.397	0.71	**0.70**
0.00075	**0.337**	0.397	0.73	**0.71**
0.0018	**0.381**	0.398	0.75	**0.71**
0.0042	0.460	**0.399**	0.83	**0.71**
0.01	0.566	**0.398**	0.93	**0.72**
0.024	0.691	**0.399**	0.99	**0.72**
0.056	0.796	**0.396**	0.99	**0.71**
0.13	0.889	**0.396**	1.03	**0.74**
0.32	0.939	**0.399**	1.01	**0.73**

the true model on 4,000 new patients per hospital, averaged over 100 replicates. At $\tau^2 = 0.05$ and $\epsilon = 1$, GHB-lite reduces the excess log-loss of local estimation by 49% (0.0266 to 0.0135), below the global model trained without privacy (0.0173), and no hospital is worse off on average (largest ratio to local 0.85, against 2.48 for the global model). Without privacy GHB-lite reaches 0.0106, better than the global model. At $\epsilon = 0.3$ it keeps most of its gain (0.0175 against 0.0273).

Design note. Three components of this simulation were added after the first runs failed. Without them, the method used the ridge releases without undoing the shrinkage, estimated τ^2 by DerSimonian-Laird [DerSimonian and Laird 1986] and ignored μ_0. With privacy noise at $\epsilon = 1$ its estimates of τ^2 were erratic (median 0, mean several times the truth), and it was worse than local estimation at $\tau^2 = 0.2$ and at $\epsilon = 0.3$. Each component follows the design of the paper (post-processing, Remark 4.4 and layer L3), but they were chosen with knowledge of these failures, and the results should be read accordingly. A fourth change came from Simulation 11: the releases were first centred on the anchor ($c = \mu_0$), and the shrinkage was undone with the Fisher information at μ_0. When the anchor was badly wrong, this corrupted every release, and GHB-lite without the anchor reached an excess log-loss of 0.34 against 0.06 for local estimation at an anchor error of two standard deviations. Centring on a fixed reference removes the dependence; Simulations 7, 8, 10 and 11 are reported with it. With anchor-centred releases, Simulation 7 had given a slightly smaller gain at $\epsilon = 1$ and no gain at $\epsilon = 0.3$. The first version of Simulation 6 also gave the GMRF prior the anchor μ_0, which the exchangeable prior did not have; the version reported here removes it.

Simulation 8: federated and meta-analytic baselines. The logistic world of Simulation 7, at $\epsilon = 1$ and four levels of heterogeneity. All private methods use the same releases, so they satisfy the same guarantee: one-shot federated averaging [McMahan et al. 2017] (the sample-size weighted mean of the releases, one model for every hospital), the random-effects meta-analytic mean (DerSimonian-Laird [DerSimonian and Laird 1986] weights, one model for every hospital), and a Ditto-style [Li et al. 2021] personalized model, which fits each hospital's data with a proximal term $\frac{\lambda}{2}\|\theta - \bar{\theta}_{\text{FedAvg}}\|^2$. The weight λ is chosen from $\{0.3, 1, 3, 10, 30, 100\}$ on ten separate tuning replicates per setting, which favours Ditto-style [Li et al. 2021]. The global model fitted on the pooled data without privacy is a reference that no private method can match in general. Results (Figure S8, left, and Table 3 of the paper; 40 replicates per setting): at $\tau^2 = 0.05$, GHB-lite 0.0134, Ditto-style [Li et al. 2021] 0.0156, pooled 0.0167, meta-analysis 0.0303, federated averaging [McMahan et al. 2017] 0.0365, local 0.0274. At $\tau^2 = 0.5$, GHB-lite 0.0282, Ditto-style [Li et al. 2021] 0.0265, local 0.0273. One-shot federated averaging [McMahan et al. 2017] is weak here because it averages noisy releases without

Table S8. Simulation 7: exact values behind Figure S7. Excess log-loss over the true model (10^{-3} nats) and AUC, mean over 24 hospitals and 100 replicates, and the largest ratio of any hospital's mean excess log-loss to its local one (Worst). Oracle: the true model of each hospital. Global: one model on the pooled data, without privacy.

		Excess log-loss			AUC				Worst	
τ^2	ϵ	Local	Global	GHB-lite	Local	Global	GHB-lite	Oracle	GHB-lite	Global
0.01	1.0	28.3	**4.1**	10.1	0.715	**0.733**	0.728	0.737	0.82	**0.77**
0.05	1.0	26.6	17.3	**13.5**	0.723	0.728	**0.731**	0.743	**0.85**	2.48
0.2	1.0	26.8	60.1	**21.4**	0.749	0.717	**0.751**	0.770	**1.02**	9.26
0.5	1.0	**27.1**	125.7	28.5	**0.784**	0.688	0.781	0.804	**1.17**	20.68
0.01	∞	27.4	**4.2**	**4.2**	0.717	**0.734**	0.733	0.737	**0.53**	0.68
0.05	∞	28.1	17.0	**10.6**	0.722	0.729	**0.734**	0.743	**0.85**	2.71
0.2	∞	27.4	59.9	**19.9**	0.745	0.712	**0.748**	0.766	**1.02**	9.56
0.5	∞	27.7	125.9	**26.0**	**0.785**	0.690	**0.785**	0.805	**1.07**	19.34
0.05	0.3	27.3	**16.7**	17.5	0.725	**0.730**	**0.730**	0.745	**0.90**	2.50
0.05	3.0	27.1	16.9	**11.3**	0.722	0.728	**0.733**	0.743	**0.87**	2.69
0.05	10.0	28.2	17.0	**10.9**	0.723	0.729	**0.734**	0.743	**0.84**	2.39

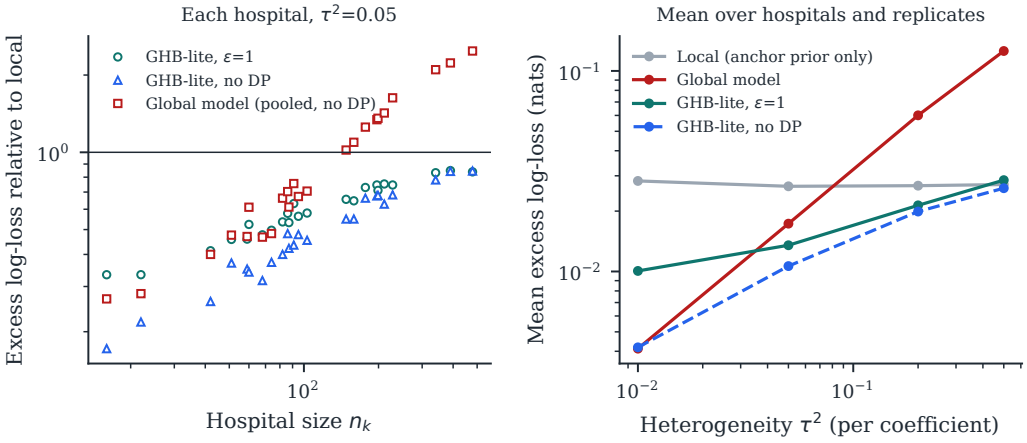


Fig. S7. Simulation 7. Left: excess log-loss relative to local estimation for each hospital ($\tau^2 = 0.05$). Right: mean excess log-loss against heterogeneity.

personalization; multi-round federated training with differentially private gradients is compared in Simulation 15.

Simulation 10: case-mix shift. As Simulation 8 at $\tau^2 = 0.05$, but each hospital draws its features from $\mathcal{N}(m_k, I)$ with $m_k \sim \mathcal{N}(0, s^2 I)$, while the de-shrinking still uses the public, unshifted feature distribution (Figure S8, right, and Table 3 of the paper). At $s = 1$: GHB-lite 0.0132, Ditto-style [Li et al. 2021] 0.0140, pooled 0.0198, local 0.0228.

Simulation 9: atypical hospitals. The network of Simulation 2 at $\tau^2 = 0.02$ and $\epsilon = 1$, with Gaussian site effects, Student-t_3 site effects of the same variance, or Gaussian effects plus one hospital at $u = 6\tau$, placed at the smallest or at the largest hospital. Three estimators: Gaussian GHB-lite; robust GHB-lite, which replaces the Gaussian prior $\mathcal{N}(\tilde{\mu}_k, \hat{\tau}^2 + M_k)$ by a Student-t_3 prior of the same variance (posterior mean by quadrature); and GHB-lite with opt-out, where a hospital keeps its local estimate when $|\hat{\theta}_k - \tilde{\mu}_k| / \sqrt{v_k + \hat{\tau}^2 + M_k} > 3$. Both variants change only the hospital's own

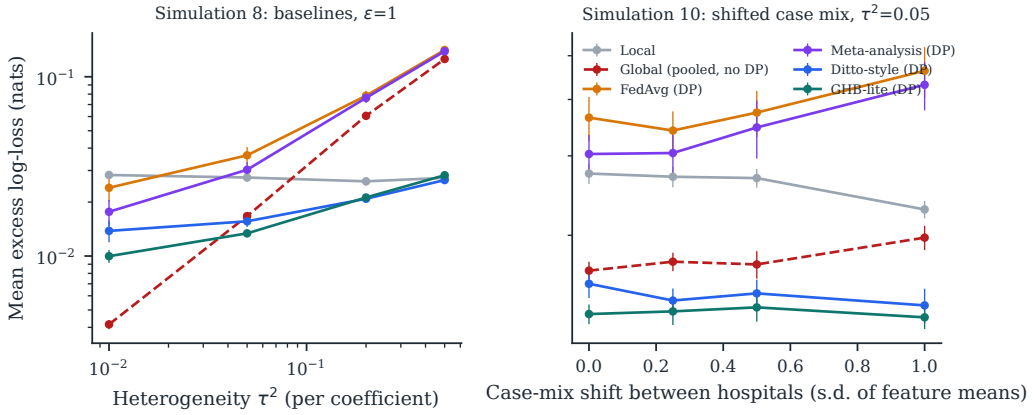


Fig. S8. Simulations 8 and 10. Mean excess log-loss of GHB-lite and of private and non-private baselines against heterogeneity (left) and case-mix shift (right). Bars: ±2 standard errors.

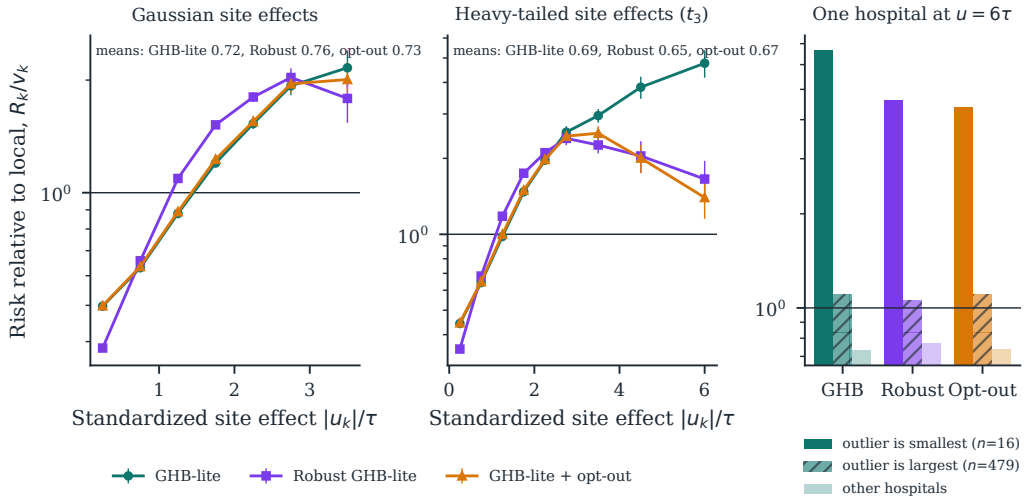


Fig. S9. Simulation 9. Left and centre: relative risk against the standardized site effect for Gaussian and heavy-tailed effects. Right: the outlying hospital and the others, when the outlier is the smallest or the largest hospital.

computation and cost no privacy. Mean relative risk: Gaussian effects 0.72 (robust 0.76), t_3 effects 0.69 (robust 0.65). The outlying hospital: 6.7 (robust 4.6, opt-out 4.4) when it is the smallest, 1.11 (robust 1.06) when it is the largest (Figure S9; Tables S9 and S10). With 16 patients, the standard error of the local estimate is as large as the deviation, so no test based on the hospital's own data can detect it reliably.

Simulation 11: a wrong anchor. The logistic world at $\tau^2 = 0.05$ and $\epsilon = 1$, with the anchor wrong by $\mathcal{N}(0, e^2 I)$ for e from 0 to 3. GHB-lite uses the anchor with its uncertainty stated honestly (e), overconfidently (0.3 whatever e), overconfidently with a conflict check (the anchor is dropped when

Table S9. Simulation 9: exact values behind Figure S9, right. Mean relative risk; with an outlier, for the outlying hospital and for the others.

	All hospitals, or the outlier			The others		
Site effects	GHB-lite	Robust	Opt-out	GHB-lite	Robust	Opt-out
Gaussian effects	**0.722**	0.763	0.728			
Student-t_3 effects	0.685	**0.653**	0.667			
outlier at the smallest hospital	6.66	4.60	**4.36**	**0.731**	0.771	0.738
outlier at the largest hospital	1.11	**1.06**	1.11	0.850	**0.748**	0.850

Table S10. Simulation 9: exact values behind Figure S9, centre. Relative risk with Student-t_3 site effects, by standardized effect $|u_k|/\tau$, with two standard errors in grey; N: hospital-replicates in the bin.

$\lvert u_k\rvert/\tau$	GHB-lite	Robust	Opt-out	N
0-0.5	0.44 (0.01)	**0.35** (0.00)	0.45 (0.01)	79,111
0.5-1	**0.64** (0.01)	0.68 (0.01)	0.65 (0.01)	38,820
1-1.5	**0.98** (0.02)	1.18 (0.02)	1.00 (0.02)	14,436
1.5-2	**1.47** (0.04)	1.75 (0.05)	1.49 (0.04)	5,717
2-2.5	**1.97** (0.08)	2.11 (0.09)	1.98 (0.08)	2,604
2.5-3	2.54 (0.14)	**2.40** (0.14)	2.45 (0.14)	1,263
3-4	2.96 (0.18)	**2.26** (0.17)	2.52 (0.16)	1,109
4-5	3.83 (0.38)	2.04 (0.29)	**2.01** (0.26)	467
5-7	4.77 (0.59)	1.66 (0.30)	**1.40** (0.25)	322

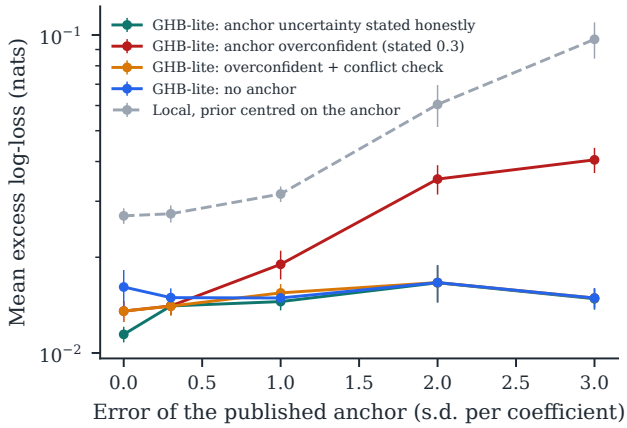


Fig. S10. Simulation 11. Mean excess log-loss against the error of the published anchor.

$|\mu_0 - \tilde{\mu}_k| > 3\sqrt{0.3^2 + M_k}$ in any coordinate), or not at all. The local model uses $\mathcal{N}(\mu_0, I)$ as its prior and degrades with the anchor (Figure S10 and Table S11). Honest uncertainty and the conflict check keep the excess log-loss at or below 0.0167 at every error, GHB-lite without the anchor at or below 0.0167; an overconfident anchor two standard deviations off reaches 0.0352, and local estimation 0.0605.

Simulation 12: dishonest hospitals. The network of Simulation 2 at $\tau^2 = 0.02$ and $\epsilon = 1$. The one or three largest hospitals release $\mu + b\tau$ instead of their privatized estimate; sizes are public, so they cannot change their weights. Robust GHB-lite obtains all releases by flooding, computes an inverse-variance weighted median and a MAD-based spread, drops releases more than three spreads

Table S11. Simulation 11: exact values behind Figure S10. Mean excess log-loss (10^{-3} nats) against the standard deviation e of the anchor's error, with two standard errors in grey; Honest: stated uncertainty e; Over.: stated 0.3; + check: stated 0.3 with the conflict check; Dropped: percentage of hospitals where the check drops the anchor.

e	Local	Honest	Over.	Over. + check	No anchor	Dropped (%)
0.0	27.0 (1.5)	**11.4** (0.6)	13.5 (1.0)	13.5 (1.0)	16.1 (2.1)	0
0.3	27.4 (1.7)	14.1 (0.9)	14.1 (0.9)	**14.0** (0.9)	14.9 (1.0)	2
1.0	31.6 (1.8)	**14.5** (0.9)	19.0 (2.0)	15.5 (1.0)	14.9 (0.9)	74
2.0	60.5 (9.1)	**16.6** (2.2)	35.2 (3.7)	16.7 (2.2)	16.7 (2.2)	100
3.0	96.9 (12.6)	**14.8** (1.1)	40.5 (3.7)	14.9 (1.1)	14.9 (1.1)	100

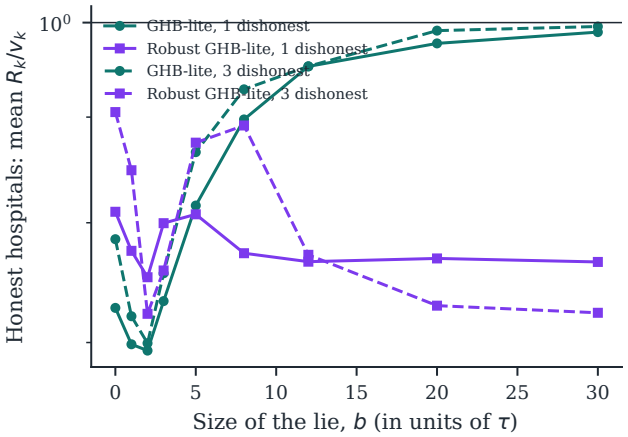


Fig. S11. Simulation 12. Honest hospitals' mean relative risk against the size of the lie, for one and three dishonest hospitals.

Table S12. Simulation 12: exact values behind Figure S11. Honest hospitals' mean relative risk against the size b of the lie (in units of τ), and the percentage of dishonest releases that robust GHB-lite keeps.

	One dishonest hospital			Three dishonest hospitals		
b	GHB-lite	Robust	Kept (%)	GHB-lite	Robust	Kept (%)
0	**0.728**	0.810	100	**0.785**	0.905	100
1	**0.699**	0.775	99	**0.721**	0.848	100
2	**0.694**	0.753	88	**0.699**	0.723	100
3	**0.733**	0.800	73	**0.756**	0.759	99
5	0.815	**0.808**	28	**0.866**	0.875	95
8	0.898	**0.773**	2	0.928	**0.892**	64
12	0.952	**0.766**	0	0.952	**0.772**	18
20	0.977	**0.769**	0	0.991	**0.729**	0
30	0.989	**0.766**	0	0.996	**0.724**	0

away, and runs GHB-lite on the rest; this is post-processing. Honest hospitals' mean relative risk (Figure S11 and Table S12): plain GHB-lite never exceeds 1.00, because a large lie inflates $\hat{\tau}^2$ and switches borrowing off; robust GHB-lite keeps 0.72 against large lies, but costs 0.81 against 0.73 when nobody lies, because the screen sometimes drops honest outlying releases.

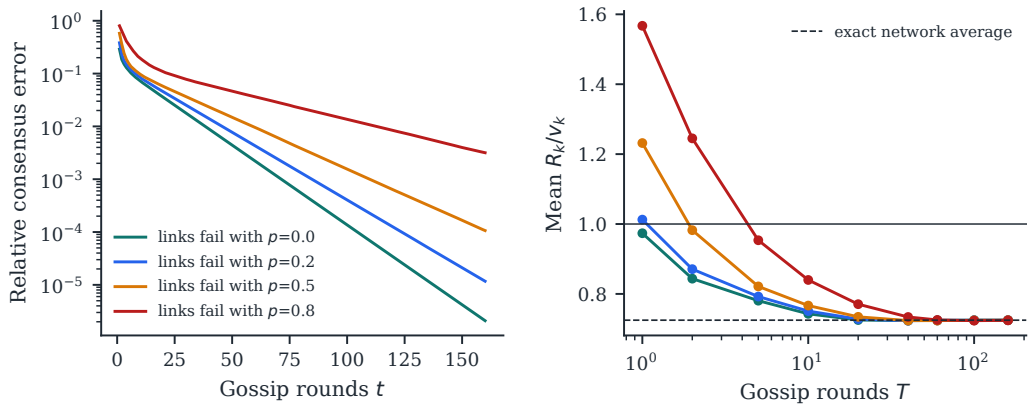


Fig. S12. Simulation 13. Left: consensus error against rounds. Right: mean relative risk of plug-in GHB-lite against rounds, for four link-failure probabilities.

Table S13. Simulation 13: exact values behind Figure S12, right. Mean relative risk of plug-in GHB-lite after T rounds, for link-failure probability p.

T	$p = 0$	$p = 0.2$	$p = 0.5$	$p = 0.8$
1	0.974	1.013	1.232	1.567
2	0.844	0.871	0.982	1.245
5	0.781	0.792	0.821	0.954
10	0.743	0.751	0.767	0.840
20	0.725	0.728	0.734	0.771
40	0.724	0.723	0.724	0.734
60	0.724	0.724	0.724	0.726
100	0.725	0.725	0.725	0.723
160	0.725	0.725	0.725	0.724
exact		0.725		

Simulation 13: link failures. The network of Simulation 2 with every link failing independently in each round with probability p; the Metropolis-Hastings weights are recomputed on the working links, so each round's matrix stays symmetric and doubly stochastic. With $p = 0.8$, the mean relative risk is 0.77 after 20 rounds and 0.73 after 60, against 0.72 with exact averaging (Figure S12 and Table S13).

Simulation 14: repeated releases. For a total budget of $\epsilon = 1$, $\delta = 10^{-5}$ spread over r refits on the same patients, the borrowed patients B after the last refit (closed form, network of Simulation 2) under basic composition ($(\epsilon/r, \delta/r)$ per release) and under zero-concentrated accounting (the Gaussian mechanism with noise s is ρ-zCDP with $\rho = \Delta^2/(2s^2)$; costs add; ρ-zCDP implies $(\rho + 2\sqrt{\rho \ln(1/\delta)}, \delta)$-DP). After 64 refits, $B = 1.5$ under basic composition and 32.4 under zCDP, against 46.1 for a single release (Figure S13 and Table S14).

Simulation 15: iterative federated learning. The logistic world of Simulation 8 (24 hospitals, ring with two-hop links, $\delta = 10^{-5}$), GHB-lite against multi-round training at the same privacy budget per hospital. GHB-lite is the revised version of Section S8; the earlier version is reported alongside it. Per-patient gradients of the logistic loss are $(p - y)x$ with $\|x\| \le \sqrt{5}$, so they need no clipping, and

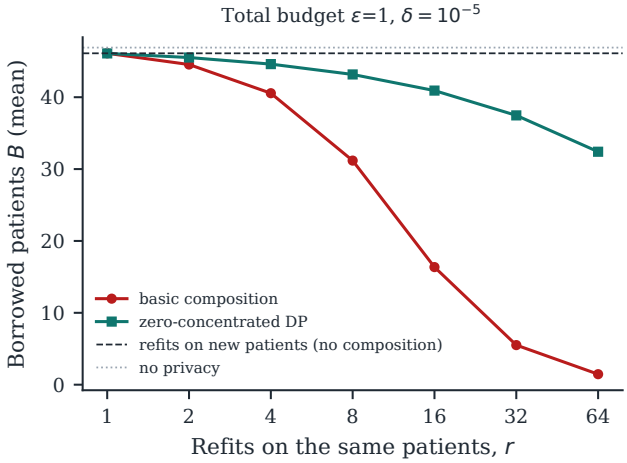


Fig. S13. Simulation 14. Borrowed patients after r refits on the same patients, for a fixed total budget.

Table S14. Simulation 14: exact values behind Figure S13. Borrowed patients B after r refits for a total budget $\epsilon = 1$, $\delta = 10^{-5}$, under basic composition and zero-concentrated accounting, and the ratio of the per-release noise standard deviations (basic over zCDP).

r	B, basic	B, zCDP	Noise ratio
1	**46.1**	**46.1**	0.99
2	44.5	**45.5**	1.44
4	40.6	**44.6**	2.09
8	31.2	**43.2**	3.03
16	16.4	**40.9**	4.40
32	5.5	**37.5**	6.36
64	1.5	**32.4**	9.20
no privacy		$B = 46.9$	

replacing one patient moves a hospital's mean gradient by at most $2\sqrt{5}/n_k$. Every iterative method adds Gaussian noise to each hospital's mean gradient in every round and is accounted with zero-concentrated DP: with R rounds, each round spends ρ/R, and the total ρ satisfies $\rho+2\sqrt{\rho \ln(1/\delta)} = \epsilon$. GHB-lite's single release is accounted in the same way. All methods start from, or shrink toward, the published anchor μ_0, which GHB-lite also uses.

- *DP-FedAvg* [McMahan et al. 2017, 2018], full-batch: a server holds one model; in each round every hospital sends its noisy mean gradient at that model, and the server takes a step of size η along the sample-size weighted average.
- *DP-FedAvg [McMahan et al. 2018] with fine-tuning*: each hospital then takes E plain gradient steps (step size 1) on its own data, starting from the global model; this costs no privacy because the data stay local.
- *Decentralized DP-SGD* [Lian et al. 2017], the algorithm behind swarm learning [Warnat-Herresthal et al. 2021]: no server; each hospital steps along its own noisy gradient and averages models with its neighbours using the Metropolis-Hastings weights of GHB-lite's graph (adapt, then combine), followed by optional fine-tuning.

- *Clustered DP-FedAvg*, in the style of IFCA [Ghosh et al. 2020]. IFCA [Ghosh et al. 2020] lets each hospital choose the cluster whose model fits its data best in every round. Under DP this choice has sensitivity of order 1, not $1/n_k$, because one patient can move a hospital's whole gradient from one cluster to another. Hospitals therefore make one private ridge release with half the budget, the releases are clustered by weighted k-means (post-processing), and DP-FedAvg [McMahan et al. 2018] runs within each cluster with the other half, followed by optional fine-tuning.
- *Local gradient descent from the anchor*, a control that uses no other hospital: E gradient steps on the hospital's own data from μ_0, with early stopping as the only regularizer.
- *Ditto-style [Li et al. 2021]* as in Simulation 8, and *GHB-lite with the true* τ^2 (the marginal heterogeneity, regional plus within-region in the regional world), a reference that no method can use.

Hyperparameters are chosen on 20 separate replicates per setting by the mean excess log-loss, which requires access to the true models: rounds $R \in \{1, 2, 3, 5, 10, 30, 100\}$ ($\{3, 10, 30, 100\}$ for decentralized training), step size $\eta \in \{0.2, 0.5, 2\}$, fine-tuning steps $E \in \{0, 1, 2, 3, 5, 10, 30, 100\}$ and clusters $C \in \{2, 3, 4\}$. Table S16 lists the choices. The only choices at the edge of a grid are $R = 1$, the smallest meaningful number of rounds, and $\eta = 0.2$ for decentralized training, where smaller steps would approach the anchor-only control. GHB-lite has no tuning parameter. Settings: $\tau^2 \in \{0.01, 0.05, 0.2, 0.5\}$ at $\epsilon = 1$; $\epsilon \in \{0.3, \infty\}$ at $\tau^2 = 0.05$; a regional world in which the 24 hospitals form four regions, with between-region variance 0.2 and within-region variance 0.02 per coefficient, where clustering should help; an anchor wrong by $\mathcal{N}(0, I)$ instead of $\mathcal{N}(0, 0.3^2 I)$; $\tau^2 = 0.1$ at $\epsilon = 0.5$; and a network of 48 hospitals (sizes drawn as for the 24, ring with two-hop links). The last four settings (marked $*$) were not used while revising GHB-lite; the development runs used other seeds, an anchor error of two standard deviations and a three-region world. Results are means over 60 further replicates (Table S15, Figure S14); Table 5 of the paper gives the worst hospital, the share of hospitals that gain, and the largest hospitals.

GHB-lite is the most accurate private method in six of the ten settings ($\tau^2 = 0.01$ and 0.05, $\epsilon = 0.3$, and the held-out regional, $\tau^2 = 0.1/\epsilon = 0.5$ and 48-hospital settings). The tuned Ditto-style [Li et al. 2021] model, which shrinks toward the average of GHB-lite's own releases, is slightly more accurate at $\tau^2 = 0.2$, without privacy and with a wrong anchor; at $\tau^2 = 0.5$ no private method beats local estimation. GHB-lite is more accurate than DP-FedAvg [McMahan et al. 2018] with fine-tuning, decentralized DP-SGD [Lian et al. 2017] and clustered training in every private setting. At $\epsilon = 0.3$, the local control that starts from the anchor reaches 0.0205 against 0.0175 for GHB-lite, and with a wrong anchor it becomes worse than local estimation (0.0419 against 0.0346). GHB-lite's worst hospital is at most 1.06 times its local error in every private setting except $\tau^2 = 0.5$ (1.15), whereas fine-tuning's worst hospital is 1.39 to 2.96 times worse than alone; the hospitals that lose are the large ones. GHB-lite with the true τ^2 remains better still (0.0158 at $\epsilon = 0.3$), so estimating heterogeneity from noisy releases is what is left to improve. The earlier version of GHB-lite reached 0.0259 at $\epsilon = 0.3$ and 0.0166 at $\tau^2 = 0.05$.

Simulation 16: ablation. The logistic world of Simulation 15 (24 hospitals, ring with two-hop links, $T = 60$, $\delta = 10^{-5}$), in eight settings: $\tau^2 \in \{0.01, 0.05, 0.2, 0.5\}$ at $\epsilon = 1$; $\epsilon \in \{0.3, \infty\}$ at $\tau^2 = 0.05$; an anchor wrong by $\mathcal{N}(0, I)$; and a case-mix shift of one standard deviation. In every replicate, every variant uses the same data, the same public sample and the same standard-normal noise draw, scaled to its own noise level, so that differences between variants are paired. Each variant switches one component of GHB-lite (`simulations/ghb_variants.py`, which reproduces the reported estimator to 10^{-10} with its default settings): the release (the sufficient statistic $X^\top y$, or a ridge fit with $\lambda = 0.2$ centred on zero and undone at the leave-one-out point, instead of the score at the anchor); the second linearization; the rule for τ (plug-in posterior mean, posterior

Table S15. Simulation 15. Mean excess log-loss (nats) over the hospitals and 60 replicates; lowest of the methods available to a hospital in bold (the earlier version of GHB-lite, the oracle and the pooled model are references). FT: fine-tuning. ∗: setting not used while revising GHB-lite.

Setting	Local	Anchor GD	GHB-lite	Ditto-style [Li et al. 2021]	DP-FedAvg [McMahan et al. 2018]	DP-FedAvg + FT [McMahan et al. 2018]	Decentr. DP-SGD [Lian et al. 2017]	Clustered DP-FedAvg [Ghosh et al. 2020]	GHB-lite, earlier	GHB-lite, true τ^2	Pooled, no DP
$\tau^2 = 0.01$	0.0284	0.0166	**0.0101**	0.0122	0.0156	0.0114	0.0166	0.0163	0.0155	0.0081	0.0043
$\tau^2 = 0.05$	0.0271	0.0190	**0.0138**	0.0154	0.0274	0.0153	0.0191	0.0214	0.0166	0.0124	0.0173
$\tau^2 = 0.2$	0.0267	0.0244	0.0217	**0.0205**	0.0732	0.0232	0.0244	0.0244	0.0207	0.0194	0.0610
$\tau^2 = 0.5$	**0.0260**	0.0293	0.0275	0.0260	0.1366	0.0287	0.0294	0.0294	0.0273	0.0240	0.1244
$\epsilon = 0.3$	0.0269	0.0205	**0.0175**	0.0306	0.0452	0.0196	0.0248	0.0206	0.0259	0.0158	0.0170
no DP	0.0278	0.0190	0.0104	**0.0101**	0.0167	0.0113	0.0118	0.0138	0.0130	0.0097	0.0167
regional*	0.0287	0.0265	**0.0209**	0.0212	0.0631	0.0249	0.0264	0.0271	0.0219	0.0199	0.0485
wrong anchor*	0.0346	0.0419	0.0162	**0.0148**	0.0446	0.0210	0.0418	0.0463	0.0194	0.0150	0.0169
$\tau^2 = 0.1$, $\epsilon = 0.5$*	0.0269	0.0228	**0.0184**	0.0238	0.0526	0.0217	0.0239	0.0229	0.0227	0.0171	0.0327
$K = 48$*	0.0324	0.0224	**0.0144**	0.0146	0.0258	0.0160	0.0225	0.0241	0.0170	0.0132	0.0172

Table S16. Simulation 15. Hyperparameters chosen on the tuning replicates: rounds R, step size η, fine-tuning steps E, clusters C, and the proximal weight λ of the Ditto-style [Li et al. 2021] model.

Setting	DP-FedAvg [McMahan et al. 2018] + FT	Decentralized DP-SGD [Lian et al. 2017]	Clustered DP-FedAvg [Ghosh et al. 2020]	Anchor GD	Ditto [Li et al. 2021] λ
$\tau^2 = 0.01$	R=3, η=2.0, E=2	R=3, η=0.2, E=5	C=2, R=1, η=0.5, E=5	E=5	10
$\tau^2 = 0.05$	R=10, η=0.5, E=5	R=3, η=0.2, E=5	C=2, R=5, η=0.2, E=5	E=5	10
$\tau^2 = 0.2$	R=2, η=2.0, E=10	R=3, η=0.2, E=10	C=3, R=1, η=0.5, E=10	E=10	3
$\tau^2 = 0.5$	R=30, η=0.2, E=10	R=3, η=0.2, E=10	C=2, R=1, η=0.5, E=10	E=10	1
$\epsilon = 0.3$	R=3, η=0.5, E=5	R=3, η=0.2, E=10	C=2, R=1, η=0.2, E=5	E=5	1
no DP	R=100, η=2.0, E=3	R=100, η=0.2, E=3	C=2, R=30, η=0.5, E=3	E=5	30
regional*	R=30, η=0.5, E=10	R=3, η=0.2, E=10	C=2, R=5, η=0.2, E=10	E=10	3
wrong anchor*	R=100, η=0.2, E=5	R=30, η=0.2, E=10	C=4, R=1, η=2.0, E=30	E=30	10
$\tau^2 = 0.1$, $\epsilon = 0.5$*	R=5, η=0.5, E=5	R=3, η=0.2, E=10	C=2, R=2, η=0.2, E=10	E=10	3
$K = 48$*	R=3, η=2.0, E=3	R=3, η=0.2, E=5	C=2, R=10, η=0.2, E=5	E=5	10

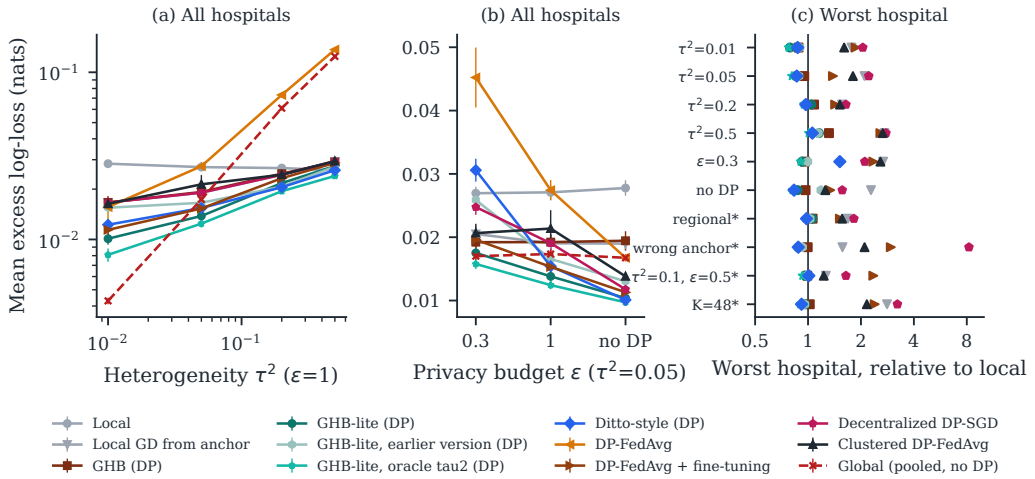


Fig. S14. Simulation 15. (a) Mean excess log-loss against heterogeneity at $\epsilon = 1$. (b) Against the privacy budget at $\tau^2 = 0.05$. (c) The worst hospital's error relative to its own local estimate; right of 1, some hospital is worse off than alone.

median or DerSimonian-Laird [DerSimonian and Laird 1986], or the posterior quantiles without weighting by the hospital's own evidence); the conflict check; the anchor; the default prior; inverse-variance weights in the network mean; gossip against exact averaging; and zero-concentrated against classical calibration of the noise. GHB-lite with the true τ^2, the earlier version of GHB-lite and local estimation are references. Table 6 of the paper gives the mean excess log-loss of each

Table S17. Simulation 16, part B: the Gaussian model. Mean relative risk R_k/v_k over the 24 hospitals of Simulation 2 (20,000 replicates per setting). FB: τ integrated out with the hospital's own evidence (reported); Mean, DL: plug-in posterior mean and DerSimonian-Laird [DerSimonian and Laird 1986]; No ev.: components without own evidence; Equal: equal weights; Exact: exact averaging; $T = 10$: gossip with 10 rounds; Oracle: τ^2 known.

τ^2	ϵ	FB	Mean	DL	No ev.	Equal	Exact	$T = 10$	Oracle
0.001	0.3	0.358	0.423	0.342	0.338	0.657	0.358	0.398	0.259
0.02	0.3	0.741	0.735	0.908	0.771	0.786	0.741	0.752	0.689
0.2	0.3	0.955	0.958	0.970	0.971	0.955	0.955	0.956	0.944
0.001	1	0.214	0.230	0.211	0.202	0.310	0.214	0.236	0.168
0.02	1	0.710	0.709	0.720	0.717	0.716	0.710	0.716	0.674
0.2	1	0.949	0.949	0.954	0.949	0.949	0.949	0.950	0.943
0.001	∞	0.189	0.195	0.186	0.178	0.209	0.188	0.206	0.153
0.02	∞	0.696	0.695	0.697	0.696	0.697	0.696	0.701	0.671
0.2	∞	0.947	0.946	0.950	0.947	0.947	0.947	0.948	0.942

variant relative to GHB-lite over 80 replicates per setting. Part B repeats the ablation of the rule for τ, the weights and the averaging in the Gaussian model of Simulation 2 over 20,000 replicates per setting (Table S17).

Simulation 17: sensitivity. Part A varies each setting of GHB-lite, the others at their reported values, in three base settings of the logistic world ($\tau^2 = 0.05$ at $\epsilon = 1$ and 0.3; $\tau^2 = 0.2$ at $\epsilon = 1$), with every value computed on the same data and releases (40 replicates): the number of posterior quantiles of τ, $Q \in \{1, 3, 6, 12, 24\}$ ($Q = 1$ is the plug-in posterior median); the half-Cauchy scale of the prior on τ, $\{0.1, 0.25, 0.5, 1, 2.5\}$; the standard deviation of the default prior, $\{1, 2.5, 5, 10\}$ or none; the conflict threshold $z \in \{2, 2.5, 3, 4, 6\}$; the stated standard deviation of the anchor, $\{0.1, 0.3, 0.6, 1, 2\}$, whose true error is 0.3; the number of gossip rounds, $T \in \{5, 10, 20, 40, 60, 120\}$; and the size of the public sample for the moments, {2,000, 5,000, 20,000, 100,000}. Ranges of the ratio to the reported settings, over all values and base settings: Q 0.98 to 1.06; scale 0.90 to 1.17; default prior 0.90 to 1.03; threshold 1.00 to 1.04; stated anchor uncertainty 1.00 to 1.41; rounds 1.00 to 1.18; public sample 1.00 to 1.00. Part B varies, at $\tau^2 = 0.05$ and $\epsilon = 1$, the number of hospitals ($K \in \{6, 12, 24, 48, 96\}$, sizes drawn as in Simulation 2), the median hospital size ($\{25, 50, 100, 200, 400\}$), the budget ($\epsilon \in \{0.1, 0.2, 0.3, 0.5, 1, 2, 5, \infty\}$, with zero-concentrated calibration, which is valid for every ϵ), $\delta \in \{10^{-3}, 10^{-5}, 10^{-7}, 10^{-9}\}$ and the four topologies of Figure 2 at $K = 24$, each with its own seeds and 40 replicates. GHB-lite's error relative to local estimation ranges from 0.40 to 0.69, and its worst hospital stays within a factor 1.03 of its local error. Figures 11 and 12 of the paper show every point, and Tables 7 and 8 give the values.

Simulations 16B and 17B: GHB. The ablation and sensitivity analysis of Simulations 16 and 17 repeated for GHB as released (`simulations/exp16b_ghb.py` and `exp17b_ghb.py`), in the same settings and conditions, with the published anchor (stated standard deviation 0.3) as in Simulation 15 and without the step across networks, since a simulated network is alone. Every variant of a replicate uses the same data and the same privacy noise (one seed per replicate, drawn from the data). Ablation: without the non-borrowing components; without the propagated privacy covariance;

Table S18. Simulation 16B: ablation and sensitivity of GHB as released (40 replicates per setting). First row: mean excess log-loss of GHB (10^{-3} nats). Other rows: the mean excess log-loss of each variant divided by that of GHB on the same data and privacy noise; bold where the paired difference exceeds two standard errors. Last row: local estimation.

Variant	$\tau^2 = 0.01$	$\tau^2 = 0.05$	$\tau^2 = 0.2$	$\tau^2 = 0.5$	$\epsilon = 0.3$	no DP	wrong anchor	shift
GHB as released (10^{-3} nats)	15.6	18.6	24.1	27.9	19.8	18.8	20.0	16.0
Without the non-borrowing components	**0.93**	**0.95**	**0.97**	0.99	**0.96**	**0.95**	**0.95**	**0.94**
Without propagating the privacy covariance	**1.00**	**1.00**	**1.00**	**1.00**	**1.01**	1.00	**1.00**	1.00
Without removing the hospital's own data	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**0.99**
Intercept borrowed like the slopes	**0.72**	**0.83**	**0.97**	1.01	**0.84**	**0.84**	**0.82**	**0.84**
Loose sensitivity of the information release	1.00	1.00	1.00	1.00	1.01	1.00	**1.00**	1.00
Full information matrix always	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Diagonal information always	1.02	0.99	1.00	1.00	1.01	**1.00**	0.98	**1.06**
Without the anchor	**1.01**	**1.01**	**1.00**	**1.00**	**1.06**	**1.00**	1.00	**1.01**
Scale of τ 0.1	**0.87**	**0.97**	**1.09**	**1.10**	**0.96**	**0.97**	**0.96**	**0.96**
Scale of τ 0.25	**0.92**	**0.96**	**1.02**	**1.04**	**0.96**	**0.97**	**0.96**	**0.96**
Scale of τ 1	**1.12**	**1.07**	1.01	**0.98**	**1.07**	**1.07**	**1.07**	**1.08**
Scale of τ 2	**1.24**	**1.16**	**1.04**	**0.99**	**1.15**	**1.15**	**1.16**	**1.17**
Non-borrowing weight 0.1	**0.96**	**0.97**	**0.98**	**0.99**	**0.97**	**0.97**	**0.96**	**0.96**
Non-borrowing weight 1	**1.09**	**1.07**	**1.03**	**1.01**	**1.06**	**1.06**	**1.07**	**1.08**
Non-borrowing weight 3	**1.24**	**1.18**	**1.08**	**1.03**	**1.16**	**1.17**	**1.18**	**1.20**
Information share 0.25	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Information share 0.75	**1.01**	1.00	1.00	1.00	**1.03**	1.00	**1.01**	1.00
$Q = 1$	**1.14**	1.01	**0.94**	0.99	0.99	1.01	1.00	1.00
$Q = 3$	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
$Q = 6$	**1.00**	1.00	1.00	1.00	**1.00**	**1.00**	**1.00**	**1.00**
$Q = 24$	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
Default scales {2.5}	1.02	1.01	**1.03**	**1.02**	1.02	1.01	1.02	1.01
Default scales {1}	**0.96**	**0.97**	**0.96**	**0.96**	**0.98**	**0.97**	**0.97**	**0.96**
Default scales {0.5, 1, 2.5, 5}	1.03	1.03	1.02	**1.02**	1.01	1.02	1.03	1.01
Local	**1.75**	**1.44**	**1.08**	**0.94**	**1.39**	**1.44**	**1.77**	**1.41**

without removing the hospital's own data; intercept borrowed like the slopes; the loose (triangle-inequality) sensitivity of the information release; the full matrix or the diagonal always instead of the public rule; without the anchor. Sensitivity: half-Cauchy scale of τ in $\{0.1, 0.25, 1, 2\}$, non-borrowing weight in $\{0.1, 1, 3\}$ (released $\frac{1}{3}$), budget share of the information release in $\{0.25, 0.75\}$, $Q \in \{1, 3, 6, 24\}$, and default scales $\{2.5\}$, $\{1\}$ or $\{0.5, 1, 2.5, 5\}$. Table S18 gives every value behind Figure 13 of the paper; Figure S15 and Table S19 give GHB and GHB-lite across network conditions, on the same worlds.

Simulation 18: group fairness. The logistic world of Simulation 7 ($\epsilon = 1, \delta = 10^{-5}$, 24 hospitals, ring with two-hop links, $T = 60$) with features $x = (1, S, z_1, z_2, z_3)$: $S \in \{0, 1\}$ is a protected attribute ($S = 1$ the minority group) and $z \sim \mathcal{N}(0, I_3)$ is clipped to $\|z\| \le \sqrt{3}$, so that $\|x\| \le \sqrt{5}$ and the sensitivity of the release is unchanged. Hospital k's minority share is $\pi_k \sim \text{Beta}(6\bar{\pi}, 6(1-\bar{\pi}))$, clipped to $[0.02, 0.98]$, with $\bar{\pi} = 0.3$ (0.1 in the rare-minority scenario); the public feature distribution used for the inversion has share $\bar{\pi}$, so hospitals whose share differs are under case-mix shift. $\theta_k = \mu + u_k$ with $\mu = (-0.5, \mu_S, 0.8, -0.6, 0.4)$, $u_k \sim \mathcal{N}(0, \text{diag}(\tau^2, \tau_S^2, \tau^2, \tau^2, \tau^2))$, $\tau^2 = 0.05$, and $(\mu_S, \tau_S^2) = (0.5, 0.05)$ (shared), $(0.5, 0.5)$ (varies), $(0.5, 0.05)$ with $\bar{\pi} = 0.1$ (rare) or $(0, 0.5)$ (opposite); the anchor is μ plus $\mathcal{N}(0, 0.3^2 I)$. *GHB-lite, S local* replaces, in every mixture component of each hospital's prior, the network estimate of the coefficient of S and its variance by the local prior $\mathcal{N}(\mu_{0,S}, 1)$. FAIR-FATE follows Algorithm 1 of [Salazar et al. 2023] with the settings in the caption of Table 9 of the paper; the fairness of a model on the server's validation set is its SP, EO or EQO ratio at threshold 0.5, in the form $\min(a, b)/\max(a, b)$, and EQO averages the ratios of true and false positive rates.



Fig. S15. Simulation 17B: GHB as released and GHB-lite across network conditions, on the same worlds ($\tau^2 = 0.05$, $\epsilon = 1$, 24 hospitals of median size 100 on the ring with two-hop links unless varied; 40 replicates per point). Mean excess log-loss relative to local estimation (±2 standard errors) and the largest ratio of any hospital's error to its own local error for GHB (dotted).

Table S19. Simulation 17B: exact values behind Figure S15. Mean excess log-loss of GHB and GHB-lite relative to local estimation, the largest ratio of any hospital's error to its own local error under GHB, and the absolute mean excess log-loss of GHB and of local estimation (10^{-3} nats).

Condition	Value	GHB / local	GHB-lite / local	Worst (GHB)	GHB	Local
Hospitals	6	0.716	0.590	0.94	17.3	24.1
Hospitals	12	0.708	0.554	0.96	18.7	26.5
Hospitals	24	0.677	0.503	1.11	18.6	27.5
Hospitals	48	0.672	0.448	0.98	21.8	32.4
Hospitals	96	0.701	0.449	1.00	20.5	29.2
Median size	25	0.745	0.585	0.89	46.8	62.8
Median size	50	0.642	0.536	0.84	28.3	44.2
Median size	100	0.677	0.509	0.95	18.1	26.7
Median size	200	0.705	0.541	1.01	11.3	16.0
Median size	400	0.820	0.671	1.06	6.8	8.2
ϵ	0.1	0.770	0.684	0.99	21.3	27.7
ϵ	0.2	0.736	0.663	0.93	19.9	27.0
ϵ	0.3	0.714	0.632	0.95	19.0	26.6
ϵ	0.5	0.733	0.587	0.96	19.3	26.4
ϵ	1	0.713	0.503	0.97	20.3	28.4
ϵ	2	0.691	0.427	0.96	19.2	27.8
ϵ	5	0.687	0.429	0.93	19.2	27.9
ϵ	no DP	0.695	0.393	0.92	19.3	27.8
δ	0.001	0.692	0.474	0.91	18.7	27.1
δ	1e-05	0.655	0.509	1.04	18.1	27.6
δ	1e-07	0.711	0.531	1.00	19.5	27.5
δ	1e-09	0.712	0.535	0.96	18.7	26.2
Topology	Ring (degree 2)	0.699	0.502	0.92	19.6	28.1
Topology	Regional clusters + bridges	0.717	0.495	0.91	20.7	28.9
Topology	Sparse random (p=0.15)	0.691	0.490	0.90	19.2	27.8
Topology	Ring + 2-hop (degree 4)	0.685	0.500	1.03	19.4	28.3

The validation set holds a quarter as many patients as each hospital's training data, drawn from that hospital's distribution, so a fifth of all patients, as in the paper. Each method is evaluated on 4,000 new patients per hospital, drawn from the hospital's own distribution; the fairness ratios

Table S20. TCGA: the literature methods, the size of each grid, and the configuration with the lowest mean error over the twelve design networks, fixed before the held-out runion (`dev/v4_realistic_configs.json`). Every method spends the same budget, with private gradients under zero-concentrated accounting.

Method	Grid	$\epsilon = 1$	$\epsilon = 10$
DP-FedAvg + FT [McMahan et al. 2018]	36	rounds R=1, step η=0.2, fine-tuning steps=5	rounds R=10, step η=0.2, fine-tuning steps=5
DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	18	rounds R=3, step η=0.5, fine-tuning steps=5	rounds R=10, step η=2.0, fine-tuning steps=5
APFL [Deng et al. 2020]	18	rounds R=3, step η=0.5, mixing α=0.75	rounds R=10, step η=0.5, mixing α=0.5
Ditto [Li et al. 2021]	18	rounds R=3, step η=0.5, proximal λ=1.0	rounds R=3, step η=0.5, proximal λ=1.0
pFedMe [T. Dinh et al. 2020]	12	rounds R=3, step η=2.0, proximal λ=10.0	rounds R=3, step η=2.0, proximal λ=10.0
Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018]	54	clusters=2, rounds R=1, step η=0.2, fine-tuning steps=5	clusters=2, rounds R=1, step η=0.5, fine-tuning steps=5

are computed on the pooled test patients of all hospitals, each scored by the model its hospital uses. Results over 100 replicates per scenario are in Table 9 and Figure 14 of the paper. The code is `simulations/exp18_fairness.py`.

S2 REAL DATA: TCGA DESIGN, ABLATION AND HELD-OUT NETWORKS

Data. Open-access clinical data of The Cancer Genome Atlas [Cancer Genome Atlas Research Network et al. 2013; Grossman et al. 2016] from the NCI Genomic Data Commons (https://api.gdc.cancer.gov/cases, script `simulations/fetch_tcga.py`, files in `data/tcga_v3/` and `data/tcga_v4/`), downloaded with the fields vital status, days to death, age at index, sex at birth, days to last follow-up, AJCC pathologic stage, T and N, FIGO stage, prior malignancy, primary-diagnosis flag and the tissue source site. A hospital is a tissue source site (by name) with at least five eligible patients. A patient is eligible with a known age, sex and pathologic stage (AJCC, or FIGO where AJCC is absent), and a two-year outcome: death within 730 days, or known survival beyond it.

Model. $x = (1, a, \text{male}, \text{stage III}, \text{stage IV}, \text{T3 or T4}, \text{N1 to N3}, \text{prior malignancy})$ with $a = (\text{age} - 62)/25$ clipped to $[-1, 1]$, so $\|x\| \le \sqrt{8}$ without fitting anything to the data. Stage letters and digits are dropped (IIIC1 is III).

Sets. Design: LUSC, HNSC, BLCA, STAD, LIHC, KIRC (the first set), and LUAD, COAD, READ, KIRP, PAAD, ESCA (the second set). Held out: every further TCGA project with at least four hospitals and at least fifteen deaths, fixed from counts alone before any model was fitted: BRCA, MESO, UCEC, OV and UCS. Of the other projects with downloadable data, ACC, PRAD and CESC have no or too few eligible hospitals, and THCA, KICH and TGCT too few deaths.

Protocol. Ten repetitions of five-fold cross-validation, stratified by outcome within each hospital, with the networks of a set fitted jointly so that GHB can borrow across them; every other method is fitted per network. Budgets $\epsilon \in \{1, 10\}$, $\delta = 10^{-5}$, zero-concentrated accounting. Errors are pooled held-out log-loss over repetitions and hospitals, relative to local estimation under $\mathcal{N}(0, 2.5^2)$. The baselines and their grids are in Table S20. Script `simulations/real_v3.py` (sets `dev`, `fresh`, `confirm`), with checkpoints per repetition.

Design of GHB. GHB's joint releases were selected among about fifteen variants on the design networks (`simulations/dev/v4_dev.py`, with the folds of `real_v3.py`). Iterating the releases as Newton's method, with the information released at every iterate, was dominated by noise at

Table S21. TCGA development: twelve networks, ten repetitions; held-out log-loss relative to local estimation, mean over the networks. GHB and GHB-lite with the evidence-weighted default prior. Literature methods: tuned on the same twelve networks (development tuning, in sample, which favours them) and on each network's own held-out folds (oracle).

	$\epsilon = 1$		$\epsilon = 10$	
Method	Development	Oracle	Development	Oracle
GHB, no tuning	**0.911**	**0.911**	**0.892**	**0.892**
GHB-lite (evidence prior), no tuning	0.927	0.927	0.941	0.941
Local, evidence prior	0.927	0.927	0.927	0.927
DP-FedAvg + FT [McMahan et al. 2018]	0.934	0.920	0.927	0.910
DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	0.940	0.930	0.914	0.897
APFL [Deng et al. 2020]	0.950	0.945	0.925	0.909
Ditto [Li et al. 2021]	1.012	0.991	0.916	0.908
pFedMe [T. Dinh et al. 2020]	13.994	13.415	1.392	1.379
Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018]	0.944	0.929	0.931	0.918

Table S22. TCGA held-out networks: every literature method in the oracle arm, tuned on each network's own held-out folds; relative held-out log-loss. Compare with Table 12 of the paper.

Method	BRCA	MESO	OV	UCEC	UCS	Mean
$\epsilon = 1$						
DP-FedAvg + FT [McMahan et al. 2018]	**0.895**	**0.914**	1.003	0.964	**0.917**	**0.939**
DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	0.911	1.027	0.997	**0.962**	1.019	0.983
APFL [Deng et al. 2020]	0.918	0.987	1.008	0.997	0.986	0.979
Ditto [Li et al. 2021]	0.967	1.480	**0.978**	0.997	1.556	1.196
pFedMe [T. Dinh et al. 2020]	13.333	14.317	7.600	7.028	12.970	11.049
Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018]	0.919	0.989	1.007	0.971	0.965	0.970
$\epsilon = 10$						
DP-FedAvg + FT [McMahan et al. 2018]	0.888	**0.877**	0.981	0.964	**0.864**	0.915
DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	0.880	0.905	**0.970**	**0.925**	0.879	0.912
APFL [Deng et al. 2020]	0.882	0.888	0.989	0.957	0.878	0.919
Ditto [Li et al. 2021]	**0.878**	0.893	0.973	0.946	0.866	**0.911**
pFedMe [T. Dinh et al. 2020]	1.244	1.363	1.116	1.106	1.458	1.257
Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018]	0.892	0.922	0.996	0.965	0.876	0.930

$\epsilon = 10$, because a Newton step divides the noise by the information of rare features. Releasing the information once, at the origin, propagating the privacy covariance into each hospital's prior and borrowing across networks fixed this; Table 13 of the paper gives what each component of GHB as released contributes. One Newton step was better than two, which spend half the budget per step; learning τ from per-hospital releases at half the budget did not help; institution effects across networks changed nothing to three decimals.

Design choices. Three choices were made on the design networks (`simulations/dev/v5_dev.py`), the seven small networks and UCI Heart Disease (`dev/v5_networks.py`), and FLamby, and never on the five held-out networks. Not borrowing the intercept (an extra prior variance 2.5^2 around the network value, and a default intercept scale of 2.5 whatever the scale of the slopes) improved the design networks from 0.934 to 0.923 of the local error at $\epsilon = 1$ and from 0.913 to 0.899 at $\epsilon = 10$ (four repetitions), and Fed-Heart-Disease at $\epsilon = 10$ from 0.440 to 0.427. The tighter sensitivity of the information release is a free improvement of the same guarantee. The diagonal fallback improved the design networks at $\epsilon = 1$ (to 0.915) and Fed-TCGA-BRCA at $\epsilon = 1$, where the full matrix of 40 coefficients is mostly noise. Rejected: weighting the components by their leave-one-out predictive accuracy (better on Fed-Heart-Disease by about 0.01, worse on the design networks by 0.003 and on the small networks by 0.004); a noise-aware network estimate (better at $\epsilon = 1$, worse at $\epsilon = 10$

Table S23. Tuned against tuned, on the five held-out networks, for GHB as released (exploratory): mean held-out log-loss relative to local estimation. GHB tuned over 36 configurations, every literature method over its grid of 12 to 54. Development: the configuration with the lowest mean error over the twelve development networks, applied unchanged. Oracle: the best configuration on each network's own held-out folds. pFedMe with its corrected noise.

	$\epsilon = 1$		$\epsilon = 10$	
Method	Development	Oracle	Development	Oracle
GHB, released configuration	0.925	0.925	0.909	0.909
GHB, tuned like the others	**0.920**	**0.916**	**0.900**	**0.898**
DP-FedAvg + FT [McMahan et al. 2018]	0.976	0.939	0.960	0.915
DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	1.005	0.983	0.990	0.912
APFL [Deng et al. 2020]	1.007	0.979	0.966	0.919
Ditto [Li et al. 2021]	1.200	1.196	0.919	0.911
pFedMe [T. Dinh et al. 2020]	11.726	11.049	1.291	1.257
Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018]	1.000	0.970	0.965	0.930

by 0.009); approximating the network's information by each hospital's own; off-diagonal shrinkage of the released matrix; several damped Newton rounds; and, for images, borrowing only along leading public directions.

Not borrowing the intercept in simulation. In Simulation 15, GHB with the intercept borrowed would be more accurate than GHB as released in every setting, by a factor of 1.01 ($\tau^2 = 0.5$) to 1.41 ($\tau^2 = 0.01$): the simulated hospitals differ in base rate no more than in slopes. Letting each hospital's evidence decide, with every component present twice, intercept borrowed and not, each with half the weight (`int_mix` in `ghb_v4.py`), reaches 0.0153 at $\tau^2 = 0.05$ in 20 replicates of Simulation 15, against 0.0144 with the intercept borrowed and 0.0177 without, and keeps most of the real-data gain (0.920 and 0.901 on the design networks at $\epsilon = 1$ and 10, against 0.915 and 0.897 for GHB as released). It is not part of GHB and has not been evaluated on held-out data.

Tuning GHB. GHB as released was also tuned like the literature methods (Table S23; `simulations/dev/v5_tu` over 36 configurations: the half-Cauchy scale of τ in $\{0.1, 0.25, 0.5, 1\}$, the weight of the non-borrowing components in $\{0, \frac{1}{3}, 1\}$ and the share of the budget for the information release in $\{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ (released $(0.5, \frac{1}{3}, \frac{1}{2})$), with shared folds and privacy noise. On the five held-out networks the released configuration reaches 0.925 and 0.909 at $\epsilon = 1$ and 10; the configuration best on the design networks reaches 0.920 and 0.900, and the best on each held-out network's own folds 0.916 and 0.898.

S3 REAL DATA: THE FLAMBY BENCHMARK

Protocol. FLamby [Ogier du Terrail et al. 2022], code at commit `edacf54`; script `simulations/flamby_bench.` results `supplement/results/flamby_results.json`. FLamby's preprocessing and static splits were reimplemented in NumPy and pandas and checked against its documentation: Fed-Heart-Disease has 199/104, 172/89, 30/16 and 85/45 training and test patients in Cleveland, Hungary, Switzerland and Long Beach; Fed-TCGA-BRCA has 248/63, 156/40, 164/42, 129/33, 129/33 and 40/11 in its six regions. Twenty runs per method, at $\epsilon \in \{1, 10\}$ and without privacy, $\delta = 10^{-5}$; the mean over hospitals of each hospital's test metric, as FLamby reports it.

Models. Heart: logistic regression on FLamby's 13 features, standardized with each hospital's training statistics, divided by 3 and clipped to $[-1, 1]$, so $\|x\| \le \sqrt{14}$ with the intercept. TCGA-BRCA: an exponential survival model, the Poisson regression with per-patient log-likelihood $d\eta - te^{\eta}$, follow-up t in units of ten years clipped to $[0, 1]$; the 38 binary columns are the one-hot coding of

14 variables, a public schema, so $\|x\| \le 4$ with the intercept and the continuous age. GHB's joint releases use the score $x(d-t)$ at the origin, with sensitivity $2G$, and the information $t\,xx^\top$, with Frobenius sensitivity $\sqrt{2}\,G^2$ (the bound for positive semidefinite rank-one terms of Section 5.2); the literature methods clip each patient's gradient to norm G, which is the gradient of a convex loss, so their sensitivity arguments carry over. The new code (`simulations/ghb_glm.py`) reproduces the logistic originals exactly on every configuration.

Notes. GHB-lite was not run on TCGA-BRCA (its public-moment inversion needs a public model of follow-up and censoring); the TCGA-BRCA feature bound is 4 rather than $\sqrt{40}$, for the reason above; in the Poisson version of GHB, each maximum a posteriori fit starts from the better of the prior mean and the origin, which leaves the maximum unchanged because the objective is convex, and components whose Laplace evidence is not finite receive zero weight; the oracle is chosen separately for each metric. Fed-ISIC2019 uses FLamby's encoder (EfficientNet-B0 of `efficientnet_pytorch`, ImageNet weights from the package's GitHub release, checksum-verified); with the tabular head, its features were projected to 16 dimensions with a public random matrix, and each of the eight classes had a one-vs-rest logistic head with an eighth of the budget.

Fed-ISIC2019. Balanced accuracy (FLamby's metric), mean over the six centres and 20 runs: a hospital alone 0.300, with the evidence prior 0.303, the pooled model 0.250; GHB 0.300, 0.299 and 0.300 at $\epsilon = 1$, 10 and without privacy; Ditto [Li et al. 2021] 0.296, DP-FedAvg [McMahan et al. 2018] with fine-tuning and secure aggregation [Bonawitz et al. 2017] 0.207 and pFedMe [T. Dinh et al. 2020] 0.225 at $\epsilon = 10$. With this weak projection every method that borrows stays at the local level, and the federated-averaging methods fall below it. The image version of GHB (Section S4) was designed on this dataset.

A synthetic check of the Poisson version. On simulated Poisson networks, GHB reduced the squared error of the coefficients from 3.74 to 0.75 with homogeneous hospitals and from 3.90 to 1.63 with moderate heterogeneity, and increased it from 3.80 to 4.21 with very strong heterogeneity, where borrowing should be switched off.

S4 REAL DATA: IMAGES

Model. Scripts `simulations/ghb_image.py` (the method and the multinomial literature baselines), `isic_image.py` (design) and `camelyon_image.py` with `camelyon_slides.py` (held-out evaluation). Encoder: EfficientNet-B0 [Tan and Le 2019] from `efficientnet_pytorch`, the encoder of FLamby's ISIC baseline [Ogier du Terrail et al. 2022], with its ImageNet weights (`efficientnet-b0-355c32eb` SHA-256 checked), frozen; features are the 1,280 outputs of the global average pool. Projection: the first 32 principal components of the public images' features; each hospital standardizes the projected features with its training statistics, divides by 3 and clips to $[-1, 1]$, so $\|x\| \le G = \sqrt{33}$ with the intercept. Head: softmax over K classes, $\theta \in \mathbb{R}^{33\times K}$; a patient's score $x(e_y - p)^\top$ has Frobenius norm at most $\sqrt{2}\,G$, so a replacement moves a score sum by at most $2\sqrt{2}\,G$, the sensitivity used by GHB and every literature method. GHB: a share 0.05 of the budget releases the class counts (sensitivity $\sqrt{2}$), which start the intercepts at the centred log frequencies; one damped Newton round releases the secure sum of the scores at that point, with the curvature $N\,\mathbb{E}_{\text{pub}}[(\operatorname{diag} p - pp^\top)\otimes xx^\top]$ from the public images (N is public under replacement) and each step projected on parameters whose rows sum to zero across classes, the softmax's invariance. Each hospital then forms its leave-one-out prior from the released sum, its own score and information and the public curvature, and averages the fits under the half-Cauchy components and the non-borrowing components by their evidence, as in Section 5.2 of the paper. Every method predicts $\arg\max_k s_k - \log \pi_k$ [Menon et al. 2021], with

π the mean predicted class probability on the hospital's own training images, over the classes it has seen.

Design (ISIC 2019). FLamby's Fed-ISIC2019 [Combalia et al. 2019], its six centers and fixed splits; public images for the projection and the curvature: the 8,238 ISIC 2019 Test images, unlabeled. Variants tried: one or four Newton rounds, damping, free intercepts, borrowing along fewer public directions, the projection dimension. Chosen: one round, the class-count start, the intercept handled by the logit adjustment, all 32 directions. 10 repetitions. pFedMe's rows use its corrected noise.

Held-out evaluation (Camelyon17). Camelyon17-WILDS [Koh et al. 2021] is distributed only from `worksheets.codalab.org`, whose TLS certificate had expired when this study ran; mirrors were not reachable from the computing environment. The patches were therefore rebuilt from the organizers' release of CAMELYON17 [Bándi et al. 2019] (AWS Open Data) with the WILDS preprocessing code (commit `4726775`): the 50 slides with lesion annotations, ten per center; 96-pixel patches at level 2 on a 32-pixel grid; a patch is tumor if its central region touches a tumor polygon and normal if more than 20% of it is non-tumor tissue (Otsu threshold of the saturation at level 4); per-slide and per-center class balancing with WILDS's seed. Slides were read by HTTP range requests (`tifffile`) instead of OpenSlide from a local copy; the annotation files matched their MD5 checksums. The rebuilt metadata has 29,717, 17,452, 42,527, 64,919 and 72,726 tumor patches (and as many normal) in centers 0 to 4, against 29,718, 17,452, 42,527, 64,919 and 73,361 in the released Camelyon17-WILDS; the difference in center 4 comes from one slide with more tumor than normal tissue. Pool: 600 tumor and 600 normal patches per hospital (fixed seed); each repetition splits each hospital's pool, stratified, into 200 training and 1,000 test patches. Public images: 200 tissue patches from each of ten slides of the challenge's test set (patients 100 to 199, ten distinct patients drawn with a fixed seed), unlabeled. Encoder input: the patch resized to 224×224, ImageNet normalization. Ten repetitions, $\epsilon \in \{1, 10\}$ and no privacy, $\delta = 10^{-5}$; realistic arm `dev/v4_realistic_configs.json`, oracle arm the best configuration per family and budget by the repetition-averaged mean. Balanced accuracy and log-loss per hospital, mean over hospitals. The privacy unit is a patch.

Table S24. Camelyon17 (held-out evaluation): balanced accuracy (higher is better) and log-loss of the logit-adjusted probabilities (lower is better), mean over five hospitals and 10 repetitions. Bold: the best per column among methods that use other hospitals, oracle excluded. †: not private. ⋆: best of the six methods on the test patches. pFedMe's correctly calibrated noise makes it diverge at $\epsilon \in \{1, 10\}$.

	Balanced accuracy			Log-loss		
Method	$\epsilon = 1$	$\epsilon = 10$	no privacy	$\epsilon = 1$	$\epsilon = 10$	no privacy
Local	0.885	0.885	0.885	0.319	0.319	0.319
Local, evidence prior	0.893	0.893	0.893	0.281	0.281	0.281
Pooled, no DP†	0.887	0.887	0.887	0.294	0.294	0.294
GHB, no tuning	0.893	**0.896**	**0.896**	0.280	**0.269**	**0.269**
DP-FedAvg + FT, tuned on development [McMahan et al. 2018]	**0.894**	0.894	0.895	0.373	0.352	0.352
DP-FedAvg + FT, sec. agg., tuned on development [Bonawitz et al. 2017; McMahan et al. 2018]	0.889	0.890	0.893	0.360	0.295	0.292
APFL, tuned on development [Deng et al. 2020]	0.889	0.895	**0.896**	0.392	0.328	0.329
Ditto, tuned on development [Li et al. 2021]	0.893	0.894	0.894	**0.271**	**0.269**	**0.269**
pFedMe, tuned on development [T. Dinh et al. 2020]	0.620	0.772	0.894	11.560	1.727	0.270
Clustered DP-FedAvg, tuned on development [Ghosh et al. 2020; McMahan et al. 2018]	0.891	0.895	0.895	0.374	0.368	0.368
Best of the six, oracle⋆	0.898	0.899	0.900	0.271	0.266	0.264

Results. At $\epsilon = 10$ GHB was above a hospital alone in every repetition and above every realistic literature method, by 0.0015 to 0.0021 over the four closest (one to two paired standard errors); at $\epsilon = 1$ it was above a hospital alone and slightly below DP-FedAvg with fine-tuning; without privacy it was above a hospital alone and the pooled model; its log-loss at $\epsilon = 10$ was lower than local.

Table S25. Eight-network benchmark, $\epsilon = 1$, leave-one-network-out arm: held-out log-loss relative to local estimation, per network.

	Medicare	LUSC	HNSC	BLCA	STAD	LIHC	KIRC
Hospitals K	43	17	12	13	9	7	7
GHB, no tuning	**0.956**	**0.952**	0.942	0.937	0.962	0.929	1.003
GHB-lite, no tuning	0.998	1.000	1.000	1.002	1.006	1.000	**0.996**
GHB-lite, evidence prior, no tuning	0.966	0.959	0.941	0.933	0.966	**0.922**	1.005
GHB-lite with secure aggregation, no tuning	0.965	1.007	0.970	0.955	0.984	0.962	1.033
GHB-lite, tuned prior	0.961	0.955	0.941	**0.928**	**0.959**	0.928	0.999
Local, tuned prior‡	0.961	0.955	0.940	0.927	0.957	0.928	1.002
DP-FedAvg [McMahan et al. 2018]	1.040	1.074	0.967	1.033	1.045	1.160	1.462
DP-FedAvg + FT [McMahan et al. 2018]	0.960	0.962	0.981	1.002	0.964	1.003	1.089
DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	0.957	0.961	**0.931**	1.000	0.963	1.005	1.078
Decentralized DP-SGD [Lian et al. 2017]	0.971	0.978	0.992	1.011	0.990	1.020	1.099
Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018]	0.960	0.964	0.984	1.006	0.966	1.005	1.092
Ditto-style (one-shot) [Li et al. 2021]	1.065	1.310	1.292	1.694	1.328	1.416	1.224
Ditto [Li et al. 2021]	0.961	0.962	0.947	0.949	0.964	0.953	1.001
pFedMe [T. Dinh et al. 2020]	2.442	4.925	8.065	7.199	8.881	13.6	11.3
APFL [Deng et al. 2020]	0.957	0.966	0.961	0.976	0.986	0.971	1.101
Per-FedAvg [Fallah et al. 2020]	1.026	1.122	1.033	1.162	1.116	1.275	1.423
FedRep [Collins et al. 2021]	0.981	0.975	1.007	1.138	1.074	1.071	1.075
FedProx [Li et al. 2020]	0.970	0.997	1.002	1.047	1.018	1.044	1.010

Table S26. Eight-network benchmark, $\epsilon = 10$, leave-one-network-out arm: held-out log-loss relative to local estimation, per network.

	Medicare	LUSC	HNSC	BLCA	STAD	LIHC	KIRC
Hospitals K	43	17	12	13	9	7	7
GHB, no tuning	0.948	**0.945**	**0.928**	**0.901**	0.934	**0.919**	**0.973**
GHB-lite, no tuning	0.964	0.983	0.953	0.954	0.945	0.965	0.987
GHB-lite, evidence prior, no tuning	0.954	0.968	0.936	0.923	0.949	0.923	0.989
GHB-lite with secure aggregation, no tuning	0.955	0.974	0.935	0.925	0.947	0.930	0.983
GHB-lite, tuned prior	0.956	0.968	0.938	0.914	0.948	0.933	0.994
Local, tuned prior‡	0.961	0.955	0.940	0.927	0.957	0.928	1.002
DP-FedAvg [McMahan et al. 2018]	0.979	1.041	0.957	0.993	1.024	1.151	1.161
DP-FedAvg + FT [McMahan et al. 2018]	0.948	0.963	0.932	0.914	0.956	0.942	1.012
DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	**0.942**	0.946	0.933	0.903	**0.926**	0.981	0.988
Decentralized DP-SGD [Lian et al. 2017]	0.956	0.962	0.932	0.919	0.962	0.946	1.071
Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018]	0.957	0.962	0.931	0.926	0.962	0.962	1.070
Ditto-style (one-shot) [Li et al. 2021]	0.985	0.995	0.977	1.193	0.975	0.996	0.984
Ditto [Li et al. 2021]	0.959	0.954	0.939	0.917	0.951	0.936	0.988
pFedMe [T. Dinh et al. 2020]	0.977	1.053	1.113	1.053	1.157	1.347	1.225
APFL [Deng et al. 2020]	0.951	0.962	0.938	0.947	0.962	0.955	1.026
Per-FedAvg [Fallah et al. 2020]	0.982	1.013	0.943	0.964	1.008	1.071	1.197
FedRep [Collins et al. 2021]	0.977	0.956	0.993	1.097	1.039	1.049	1.051
FedProx [Li et al. 2020]	0.951	0.961	0.931	0.922	0.959	0.935	1.046

Over the local model with the evidence-weighted prior, GHB gained 0.0036 of balanced accuracy at $\epsilon = 10$, in all ten repetitions; Ditto (realistic) matched its log-loss.

S5 REAL DATA: EIGHT SMALLER NETWORKS

These tables give every network's result behind Table 10 of the paper, in the leave-one-network-out arm: every literature method uses the configuration with the lowest mean error over the other networks. Medicare is Arizona Medicare (43 hospitals); the TCGA networks use the five-coefficient head of the paper and are fitted one at a time. Values are held-out log-loss relative to local estimation; bold marks the lowest per network among methods that use other hospitals.

Table S27. UCI Heart Disease: hyperparameters of the baselines chosen on the held-out folds, by the maximum number of training patients per hospital.

Training patients	Local GD, tuned	DP-FedAvg [McMahan et al. 2018] + FT ($\epsilon = 1$)	Ditto-style [Li et al. 2021] ($\epsilon = 1$)
20	E=30	R=1, η=0.2, E=30	λ=0.01
40	E=30	R=3, η=0.2, E=30	λ=0.01
80	E=100	R=10, η=0.2, E=30	λ=0.01
all	E=100	R=3, η=2.0, E=30	λ=0.1

S6 REAL DATA: UCI HEART DISEASE

Data. The processed Cleveland, Hungarian, Switzerland and VA Long Beach files of the UCI Heart Disease data [Detrano et al. 1989] (CC BY 4.0), as distributed by the UCI Machine Learning Repository; they are in `data/heart_disease/` of the repository. The outcome is angiographic coronary disease (num > 0). Sizes and prevalences: Cleveland 303 (46%), Budapest 294 (36%), Zurich and Basel 123 (93%), Long Beach 200 (75%). Age, sex and chest-pain type are complete at every site; the exercise-test variables are missing for about a quarter of the Long Beach patients, and cholesterol is recorded as zero at the Swiss sites, so they are not used.

Model and privacy. $x = (1, a, \text{male}, \text{asymptomatic}, \text{anginal})$ with $a = (\text{age} - 55)/25$ clipped to $[-1, 1]$; non-anginal pain is the reference, and anginal combines typical and atypical angina. Hence $\|x\| \le 2 \le \sqrt{5}$, and the release (the mean score at 0, $\epsilon \in \{1, 5\}$, $\delta = 10^{-5}$, zero-concentrated calibration) has the sensitivity used throughout. GHB-lite is `ghb_logit.py` without an anchor: the conflict check is off, the second linearization point is the leave-one-out network estimate combined with the default prior $\mathcal{N}(0, 2.5^2)$ in place of an anchor (without it, the linearization point of a hospital with 20 patients can reach the clipping bound and make the public information matrix singular), and the final fit uses only the network components and the default prior. Gossip runs for 60 rounds on the ring of four hospitals. The public feature distribution is age $\mathcal{N}(55, 9^2)$, male with probability 0.75 and chest pain asymptomatic, anginal or non-anginal with probabilities 0.5, 0.3 and 0.2, independently: a nominal referral population, not estimated from the data. With the public information at 0, the standard deviation of the privacy noise on the parameter scale is, at the median coefficient, 26 times the sampling standard error of a hospital of 40 patients at $\epsilon = 1$, 5.6 times at $\epsilon = 5$, and 9.5 and 2.0 times for a hospital of 300 patients.

Evaluation. Twenty repeats of stratified five-fold cross-validation within each hospital, with the folds of the four hospitals aligned so that every split has one training set and one test set per hospital. The training set is subsampled, stratified by outcome, to at most 20, 40 or 80 patients, or kept whole. All methods are fitted on the training sets and scored on each hospital's own test fold by log-loss, Brier score and AUC. Baselines (`simulations/literature.py`): DP-FedAvg [McMahan et al. 2018] from 0 with $R \in \{1, 3, 10, 30\}$ rounds, step $\eta \in \{0.2, 0.5, 2\}$ and $E \in \{0, 1, 5, 30\}$ full-batch fine-tuning steps of size 1; the Ditto-style [Li et al. 2021] model, a MAP fit with prior $\mathcal{N}(\bar{\theta}, \lambda^{-1} I)$ around the size-weighted mean $\bar{\theta}$ of GHB-lite's inverted releases at $\epsilon = 1$, $\lambda \in \{0.01, 0.1, 0.3, 1, 3, 10, 30, 100\}$; and local gradient descent from 0 with step 1 and $E \in \{1, 2, 3, 5, 10, 30, 100\}$ steps. Each is tuned by its mean held-out log-loss over all splits and hospitals, separately for each training size (Table S27). Local estimation and the pooled model are MAP fits with the default prior $\mathcal{N}(0, 2.5^2)$. The code is `simulations/real_heart.py`; Table S28 gives every hospital's results.

Table S28. UCI Heart Disease: held-out log-loss, Brier score and AUC of every method at each hospital (Cle: Cleveland; Bud: Budapest; Swi: Zurich and Basel; VA: Long Beach), mean over 100 splits. †: not private.

	Log-loss				Brier score				AUC			
Method	Cle	Bud	Swi	VA	Cle	Bud	Swi	VA	Cle	Bud	Swi	VA
At most 40 training patients												
Local	0.533	0.514	0.232	0.563	**0.175**	0.165	0.063	0.186	0.820	0.814	0.750	0.644
GHB-lite, $\epsilon = 1$	**0.532**	0.514	0.232	0.563	**0.175**	0.165	0.063	0.187	**0.821**	0.813	0.745	0.645
GHB-lite, $\epsilon = 5$	0.533	0.510	0.234	0.556	0.176	0.164	0.064	0.184	**0.821**	0.818	0.738	0.659
DP-FedAvg + FT, $\epsilon = 1$ [McMahan et al. 2018]	0.542	**0.501**	**0.217**	**0.541**	0.179	**0.163**	**0.058**	**0.179**	0.819	**0.825**	**0.776**	0.666
Ditto-style, $\epsilon = 1$ [Li et al. 2021]	0.616	0.628	0.479	0.720	0.182	0.173	0.076	0.202	0.817	0.801	0.637	0.622
Local GD, tuned stopping	0.542	**0.501**	**0.217**	0.542	0.179	**0.163**	**0.058**	0.180	0.819	**0.825**	0.766	**0.668**
GHB-lite, no DP†	0.516	0.491	0.231	0.542	0.169	0.159	0.063	0.180	0.833	0.828	0.729	0.690
GHB-lite, empirical features†	0.525	0.494	0.226	0.537	0.173	0.160	0.061	0.178	0.829	0.827	0.733	0.685
Global†	0.528	0.512	0.364	0.537	0.174	0.167	0.108	0.177	0.835	0.831	0.737	0.709
All training patients												
Local	0.502	0.485	0.220	0.529	0.165	0.156	0.060	0.174	**0.841**	0.832	0.737	0.689
GHB-lite, $\epsilon = 1$	0.502	0.484	0.220	0.529	0.165	0.156	0.060	0.174	**0.841**	0.832	0.748	0.692
GHB-lite, $\epsilon = 5$	**0.501**	0.484	0.225	0.528	**0.164**	**0.155**	0.061	0.174	0.840	0.831	**0.754**	0.698
DP-FedAvg + FT, $\epsilon = 1$ [McMahan et al. 2018]	0.525	0.487	**0.211**	**0.524**	0.172	0.158	**0.057**	**0.173**	0.826	0.834	0.752	**0.706**
Ditto-style, $\epsilon = 1$ [Li et al. 2021]	0.503	0.486	0.228	0.532	0.165	0.156	0.062	0.176	0.840	0.832	0.740	0.687
Local GD, tuned stopping	0.504	**0.483**	0.213	0.526	0.165	0.156	0.058	**0.173**	0.838	**0.836**	0.725	0.697
GHB-lite, no DP†	0.501	0.481	0.224	0.526	0.164	0.155	0.061	0.174	0.838	0.833	0.750	0.699
GHB-lite, empirical features†	0.504	0.480	0.219	0.525	0.165	0.155	0.059	0.173	0.837	0.834	0.754	0.702
Global†	0.503	0.488	0.397	0.531	0.164	0.158	0.121	0.177	0.840	0.834	0.758	0.711

S7 STATISTICAL TESTS

Plan. The tests follow a plan written after every result was known and committed before any test was computed (`simulations/dev/STATS_PLAN.md`; code `simulations/stats_tests.py`, output `supplement/results/stats_tests.json`). To keep the results from steering the choice of tests, the plan uses conventional tests, the strongest available comparator and a standard multiplicity correction, and reports every outcome. The unit of evidence for a claim about networks is the network: repetitions within a network average out splits and privacy noise, but are not independent evidence about new networks. Four families of tests are run, each with its own Holm correction [Holm 1979] at a familywise level of 0.05.

(1) **Primary, confirmatory:** the five TCGA networks held out from GHB's development, network as unit. At $\epsilon = 1$ and $\epsilon = 10$: is GHB better than a hospital alone, and better than the strongest realistic competitor (the literature method, tuned on the design networks, with the lowest mean at that budget)? One-sided exact Wilcoxon signed-rank tests [Wilcoxon 1945] on the five paired differences; with five networks the smallest attainable p is $1/32 = 0.031$. The oracle arm, tuned on the test folds, is reported with an interval but not tested, since no consortium can attain it.

(2) **Confirmatory, one network:** the four claims stated before Camelyon17 was evaluated, over the 10 paired repetitions: at $\epsilon = 10$ above a hospital alone and above each realistic method; at $\epsilon = 1$ not more than 0.005 below a hospital alone (non-inferiority); without privacy above a hospital alone and above pooling. Ten one-sided exact Wilcoxon tests.

(3) **Descriptive:** all 16 real data sets (the five held-out TCGA networks, the seven small networks, FLamby Heart, TCGA-BRCA and ISIC 2019, and Camelyon17), data set as unit, each on its paper metric: one-sided exact sign tests of GHB against a hospital alone and against the strongest realistic competitor of each data set (the realistic method with the best mean on it; on the seven small networks, tuned on the other networks), at both budgets.

Ten of the sixteen data sets were used while GHB was developed, so this family describes and cannot confirm.

(4) **Descriptive:** Simulation 15, setting as unit: GHB against each method, a z statistic from the saved Monte Carlo standard errors combined as if independent (conservative when the replicates are paired), two-sided, Holm within each comparison.

A superiority claim in the paper requires an adjusted p below 0.05 in the first or second family.

Results. Table S29 gives the confirmatory families. On the held-out TCGA networks GHB was better than a hospital alone on all five networks at both budgets and better than the best realistic method on all five at $\epsilon = 1$, each with $p = 0.031$, the smallest five networks allow, but $p = 0.125$ after the Holm correction over the four primary tests; at $\epsilon = 10$ it was better than Ditto on 3 of five ($p = 0.312$). No primary hypothesis is therefore established at the familywise level of 0.05. The interval against the oracle arm at $\epsilon = 1$, [-0.024, -0.004], excludes zero, but that comparison was not tested. On Camelyon17, 7 of the 10 tests are established: GHB beats a hospital alone at every budget (non-inferiority at $\epsilon = 1$, superiority otherwise), pooling without privacy, and three of the six realistic methods at $\epsilon = 10$; its leads over DP-FedAvg with fine-tuning [McMahan et al. 2018], APFL [Deng et al. 2020] and clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018] are not. Across all sixteen data sets (Tables S30 and S31), GHB improved on a hospital alone in 14 at $\epsilon = 1$ and 15 at $\epsilon = 10$ (Holm $p = 0.006$ and 0.001) and beat the strongest realistic competitor in 11 and 10 (Holm $p = 0.210$ and 0.227). In Simulation 15 (Table S32), GHB with joint releases is significantly better than a hospital alone in 9 of the 10 settings and worse only at $\tau^2 = 0.5$, better than DP-FedAvg in 8, but significantly worse than the tuned Ditto-style model in 8 settings and than DP-FedAvg with fine-tuning in 4; in that world, whose hospitals share their base rates, GHB-lite is the variant that leads (main paper, Simulation 15).

Reading. The tests establish that GHB, untuned, improves on a hospital working alone, across the real data sets and on the held-out image network. They do not establish that it beats the tuned literature: it had the lowest mean error on the held-out cancer networks, at $\epsilon = 1$ on every one of them, but five networks are too few for that to survive a multiplicity correction, and elsewhere it leads about as often as it trails. More held-out networks are what would settle it; with ten, winning nine would give a one-sided sign-test p of 0.011 before correction.

Table S29. Confirmatory tests. Mean difference: GHB minus the comparator, relative log-loss on TCGA (negative favours GHB), balanced accuracy on Camelyon17 (positive favours GHB), with a 95% t interval over networks or repetitions. Better: networks or repetitions where GHB is better. p: one-sided exact Wilcoxon signed-rank test; Holm: adjusted within each family. Bold: established at 0.05 after adjustment. Oracle rows are reported, not tested.

Data	ε	GHB against	Mean difference [95% CI]	Better	p	Holm p
TCGA confirmation (5 networks)	1	Local	−0.075 [−0.127, −0.023]	5 of 5	0.031	0.125
TCGA confirmation (5 networks)	1	DP-FedAvg + FT [McMahan et al. 2018] (realistic)	−0.052 [−0.121, +0.018]	5 of 5	0.031	0.125
TCGA confirmation (5 networks)	1	DP-FedAvg + FT [McMahan et al. 2018] (oracle)	−0.014 [−0.024, −0.004]	5 of 5	not tested	not tested
TCGA confirmation (5 networks)	10	Local	−0.091 [−0.143, −0.040]	5 of 5	0.031	0.125
TCGA confirmation (5 networks)	10	Ditto [Li et al. 2021] (realistic)	−0.010 [−0.043, +0.023]	3 of 5	0.312	0.312
TCGA confirmation (5 networks)	10	Ditto [Li et al. 2021] (oracle)	−0.003 [−0.023, +0.017]	3 of 5	not tested	not tested
Camelyon17 (10 repetitions)	10	Local	+0.011 [+0.009, +0.013]	10 of 10	<0.001	**0.010**
Camelyon17 (10 repetitions)	10	DP-FedAvg + FT [McMahan et al. 2018] (realistic)	+0.002 [−0.001, +0.005]	7 of 10	0.116	0.349
Camelyon17 (10 repetitions)	10	DP-FedAvg-SA + FT [Bonawitz et al. 2017; McMahan et al. 2018] (realistic)	+0.006 [+0.004, +0.009]	9 of 10	0.002	**0.010**
Camelyon17 (10 repetitions)	10	APFL [Deng et al. 2020] (realistic)	+0.002 [−0.001, +0.004]	7 of 10	0.116	0.349
Camelyon17 (10 repetitions)	10	Ditto [Li et al. 2021] (realistic)	+0.002 [+0.001, +0.003]	9 of 10	0.005	**0.020**
Camelyon17 (10 repetitions)	10	pFedMe [T. Dinh et al. 2020] (realistic)	+0.124 [+0.101, +0.147]	10 of 10	<0.001	**0.010**
Camelyon17 (10 repetitions)	10	Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018] (realistic)	+0.002 [−0.002, +0.005]	6 of 10	0.161	0.349
Camelyon17 (10 repetitions)	1	Local, non-inferiority	+0.008 [+0.005, +0.010]	10 of 10	<0.001	**0.010**
Camelyon17 (10 repetitions)	∞	Local	+0.011 [+0.009, +0.013]	10 of 10	<0.001	**0.010**
Camelyon17 (10 repetitions)	∞	Pooled	+0.009 [+0.007, +0.012]	10 of 10	<0.001	**0.010**

Table S30. Descriptive tests over the sixteen real data sets: the number of data sets where GHB is better, a one-sided exact sign test (ties count against GHB) and the Holm-adjusted p over the four tests. Bold: below 0.05 after adjustment.

ε	GHB against	GHB better in	p (sign test)	Holm p
1	Local	14 of 16	0.002	**0.006**
1	strongest realistic competitor	11 of 16	0.105	0.210
10	Local	15 of 16	<0.001	**0.001**
10	strongest realistic competitor	10 of 16	0.227	0.227

Table S31. The sixteen real data sets behind Table S30, on each data set's paper metric: relative log-loss on the TCGA and small networks (lower is better), log-loss on FLamby Heart (lower), concordance on FLamby TCGA-BRCA and balanced accuracy on ISIC 2019 and Camelyon17 (higher). Strongest realistic: the realistic literature method with the best value on that data set. Bold: the better of GHB and that method. [c]: held out from GHB's development.

	$\varepsilon = 1$				$\varepsilon = 10$			
Data set	GHB	Local	Strongest realistic		GHB	Local	Strongest realistic	
TCGA-BRCA[c]	**0.879**	1.000	DP-FedAvg + FT [McMahan et al. 2018] (realistic)	0.898	**0.859**	1.000	Ditto [Li et al. 2021] (realistic)	0.878
TCGA-MESO[c]	**0.909**	1.000	DP-FedAvg + FT [McMahan et al. 2018] (realistic)	0.952	0.897	1.000	Ditto [Li et al. 2021] (realistic)	**0.893**
TCGA-UCEC[c]	**0.957**	1.000	Ditto [Li et al. 2021] (realistic)	0.997	**0.932**	1.000	DP-FedAvg-SA + FT [Bonawitz et al. 2017; McMahan et al. 2018] (realistic)	0.957
TCGA-OV[c]	0.979	1.000	Ditto [Li et al. 2021] (realistic)	**0.978**	**0.967**	1.000	Ditto [Li et al. 2021] (realistic)	0.974
TCGA-UCS[c]	**0.899**	1.000	DP-FedAvg + FT [McMahan et al. 2018] (realistic)	0.921	0.887	1.000	Ditto [Li et al. 2021] (realistic)	**0.866**
medpar	**0.956**	1.000	DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	0.957	0.948	1.000	DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	**0.942**
TCGA-LUSC	**0.952**	1.000	DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	0.961	**0.945**	1.000	DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	0.946
TCGA-HNSC	0.942	1.000	DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	**0.931**	**0.928**	1.000	Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018]	0.931
TCGA-BLCA	**0.937**	1.000	Ditto [Li et al. 2021]	0.949	**0.901**	1.000	DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	0.903
TCGA-STAD	**0.962**	1.000	DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	0.963	0.934	1.000	DP-FedAvg + FT, sec. agg. [Bonawitz et al. 2017; McMahan et al. 2018]	**0.926**
TCGA-LIHC	**0.929**	1.000	Ditto [Li et al. 2021]	0.953	**0.919**	1.000	FedProx [Li et al. 2020]	0.935
TCGA-KIRC	1.003	1.000	Ditto [Li et al. 2021]	**1.001**	**0.973**	1.000	Ditto-style (one-shot) [Li et al. 2021]	0.984
FLamby heart	0.443	0.470	Ditto [Li et al. 2021]	**0.430**	0.427	0.470	DP-FedAvg-SA + FT [Bonawitz et al. 2017; McMahan et al. 2018]	**0.419**
FLamby brca	**0.785**	0.738	DP-FedAvg-SA + FT [Bonawitz et al. 2017; McMahan et al. 2018]	0.747	0.801	0.738	DP-FedAvg-SA + FT [Bonawitz et al. 2017; McMahan et al. 2018]	**0.813**
FLamby isic	**0.300**	0.300	Ditto [Li et al. 2021]	0.296	**0.299**	0.300	Ditto [Li et al. 2021]	0.296
Camelyon17	0.893	0.885	DP-FedAvg + FT [McMahan et al. 2018]	**0.894**	**0.896**	0.885	APFL [Deng et al. 2020]	0.895

Table S32. Simulation 15: the number of settings where GHB (joint releases) is significantly better, not distinguishable or significantly worse than each method, two-sided, Holm-adjusted within each row, from the saved Monte Carlo standard errors.

GHB against	Scenarios	GHB better	Not distinguishable	GHB worse
DP-FedAvg [McMahan et al. 2018]	10	8	1	1
DP-FedAvg + fine-tuning [McMahan et al. 2018]	10	0	6	4
Decentralized DP-SGD [Lian et al. 2017]	10	2	7	1
Clustered DP-FedAvg [Ghosh et al. 2020; McMahan et al. 2018]	10	1	8	1
Ditto-style (DP) [Li et al. 2021]	10	2	0	8
Local	10	9	0	1

S8 REVISION OF GHB-LITE IN SIMULATION AND EXTENSIONS TRIED

The first run of Simulation 15 showed that the earlier GHB-lite lost to DP-FedAvg [McMahan et al. 2018] with fine-tuning when hospitals were nearly identical and when the budget was strict. A diagnosis on development seeds (900 to 904, used by no reported simulation) found two causes. First, its plug-in τ^2, the posterior mean under a half-Cauchy prior, is driven by the heavy prior tail when releases are noisy: it averaged 0.8 to 1.1 against a true 0.05 at $\epsilon = 0.3$, so GHB-lite barely borrowed. Second, its ridge releases ($\lambda = 0.2$) have five times the sensitivity of a score, and undoing the ridge to first order at a fixed reference left a bias in the releases of the largest hospitals without privacy (their squared errors were 2.5 to 9 times their stated variances). The revision makes three changes, each following the design of Section 4: a score release at the anchor, inverted by public moment

Table S33. Development runs (20 replicates each, development seeds): mean excess log-loss of the earlier GHB-lite, the revised GHB-lite, and local estimation. Settings use the logistic world of Simulation 7; "shift" is a case-mix shift of one standard deviation, "anchor" an anchor wrong by two standard deviations.

	$\tau^2 = 0.05$			$\tau^2 = 0.5$	$\tau^2 = 0.05, \epsilon = 1$	
	$\epsilon = 0.3$	$\epsilon = 1$	no DP	$\epsilon = 1$	shift	anchor
Local	0.0261	0.0289	0.0275	0.0272	0.0239	0.0581
GHB-lite, earlier version	0.0243	0.0166	0.0134	0.0268	0.0151	0.0194
GHB-lite, revised	0.0158	0.0143	0.0106	0.0269	0.0138	0.0160

matching at anchor-free or check-gated points (scope (d) of Section 4.2); τ integrated out over its posterior, weighted by each hospital's own marginal likelihood (Remark 4.4); and the anchor conflict check by default, decided once per hospital on anchor-free information. Table S33 gives the development runs that chose between the alternatives; the scripts are in `simulations/dev/`.

Alternatives rejected on the development runs. Releasing $X^\top y/n$ instead of the score: calibrated, but about twice the sampling variance for the slopes, because it ignores the hospital's own features. Ridge releases centred on the anchor ($\lambda = 1$ or 0.2) or on zero: competitive when the anchor is right, but two to three times worse with a wrong anchor (0.032 to 0.045 against 0.016). Plug-in alternatives to integrating τ out: the posterior median or the posterior mean of the precision borrowed more at small τ^2 but became worse than local estimation at $\tau^2 = 0.5$ (up to 0.041 against 0.028). An additional Student-t tail on the integrated prior changed nothing beyond Monte Carlo error. Without a weakly informative default prior, the widest components left tiny hospitals' fits unregularized and logistic separation made some diverge; with a conflict check applied separately per component, a wrong anchor re-entered through the wide components.

Extensions tried and not adopted. Three extensions proposed in Section 4.2 and the agenda were developed in the same way (`dev/ext.py`, `ext2.py`, `ext3.py`). (1) *Joint release* by groups of g consecutive hospitals through secure aggregation, each group entering borrowing as one unit whose heterogeneity is diluted by its size weights: groups of three over all hospitals lowered the error by at most 12% (0.0140 against 0.0160 at $\tau^2 = 0.01$, $\epsilon = 0.3$; 0.0125 against 0.0133 at $\tau^2 = 0.05$, $\epsilon = 1$) and never raised it beyond Monte Carlo error. It is worth its cost only where secure aggregation is available anyway. (2) *Two release rounds* under zCDP, the second score evaluated at each hospital's public network estimate: no gain in any setting (for example 0.0176 against 0.0177 at $\epsilon = 0.3$, 0.0138 against 0.0135 at $\epsilon = 1$), because splitting the budget cancels the better linearization point. (3) *A two-level prior* for public region labels (hospital within region within network), with both variance components integrated out: a small gain in a three-region world at $\epsilon = 1$ (0.0158 against 0.0164) but losses elsewhere (0.0158 against 0.0141 without regions, 0.0215 against 0.0175 at $\epsilon = 0.3$): two variance components cannot be estimated reliably from 24 noisy releases.

REFERENCES

Péter Bándi, Oscar Geessink, Quirine Manson, Marcory Van Dijk, Maschenka Balkenhol, Meyke Hermsen, Babak Ehteshami Bejnordi, Byungjae Lee, Kyunghyun Paeng, Aoxiao Zhong, et al. 2019. From Detection of Individual Metastases to Classification of Lymph Node Status at the Patient Level: The CAMELYON17 Challenge. *IEEE Transactions on Medical Imaging* 38, 2 (2019), 550-560. https://doi.org/10.1109/TMI.2018.2867350

Keith Bonawitz, Vladimir Ivanov, Ben Kreuter, Antonio Marcedone, H. Brendan McMahan, Sarvar Patel, Daniel Ramage, Aaron Segal, and Karn Seth. 2017. Practical Secure Aggregation for Privacy-Preserving Machine Learning. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. 1175-1191. https://doi.org/10.1145/3133956.3133982

Cancer Genome Atlas Research Network, John N. Weinstein, Eric A. Collisson, Gordon B. Mills, Kenna R. Mills Shaw, Brad A. Ozenberger, Kyle Ellrott, Ilya Shmulevich, Chris Sander, and Joshua M. Stuart. 2013. The Cancer Genome Atlas

Pan-Cancer Analysis Project. *Nature Genetics* 45, 10 (2013), 1113-1120. https://doi.org/10.1038/ng.2764

Liam Collins, Hamed Hassani, Aryan Mokhtari, and Sanjay Shakkottai. 2021. Exploiting Shared Representations for Personalized Federated Learning. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR, Vol. 139)*. 2089-2099.

Marc Combalia, Noel C. F. Codella, Veronica Rotemberg, Brian Helba, Veronica Vilaplana, Ofer Reiter, Cristina Carrera, Alicia Barreiro, Allan C. Halpern, Susana Puig, and Josep Malvehy. 2019. BCN20000: Dermoscopic Lesions in the Wild. *arXiv preprint arXiv:1908.02288* (2019).

Yuyang Deng, Mohammad Mahdi Kamani, and Mehrdad Mahdavi. 2020. Adaptive Personalized Federated Learning. https://doi.org/10.48550/arXiv.2003.13461 arXiv:2003.13461 [cs.LG]

Rebecca DerSimonian and Nan Laird. 1986. Meta-analysis in clinical trials. *Controlled Clinical Trials* 7, 3 (1986), 177-188. https://doi.org/10.1016/0197-2456(86)90046-2

Robert Detrano, Andras Janosi, Walter Steinbrunn, Matthias Pfisterer, Johann-Jakob Schmid, Sarbjit Sandhu, Kern H. Guppy, Stella Lee, and Victor Froelicher. 1989. International Application of a New Probability Algorithm for the Diagnosis of Coronary Artery Disease. *The American Journal of Cardiology* 64, 5 (1989), 304-310. https://doi.org/10.1016/0002-9149(89)90524-9

Alireza Fallah, Aryan Mokhtari, and Asuman Ozdaglar. 2020. Personalized Federated Learning with Theoretical Guarantees: A Model-Agnostic Meta-Learning Approach. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Andrew Gelman, Aleks Jakulin, Maria Grazia Pittau, and Yu-Sung Su. 2008. A weakly informative default prior distribution for logistic and other regression models. *The Annals of Applied Statistics* 2, 4 (2008), 1360-1383. https://doi.org/10.1214/08-AOAS191

Avishek Ghosh, Jichan Chung, Dong Yin, and Kannan Ramchandran. 2020. An Efficient Framework for Clustered Federated Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Robert L. Grossman, Allison P. Heath, Vincent Ferretti, Harold E. Varmus, Douglas R. Lowy, Warren A. Kibbe, and Louis M. Staudt. 2016. Toward a Shared Vision for Cancer Genomic Data. *New England Journal of Medicine* 375, 12 (2016), 1109-1112. https://doi.org/10.1056/NEJMp1607591

Sture Holm. 1979. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics* 6, 2 (1979), 65–70.

Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, et al. 2021. WILDS: A Benchmark of in-the-Wild Distribution Shifts. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR, Vol. 139)*. 5637-5664.

Tian Li, Shengyuan Hu, Ahmad Beirami, and Virginia Smith. 2021. Ditto: Fair and Robust Federated Learning Through Personalization. In *Proceedings of the 38th International Conference on Machine Learning (ICML) (PMLR, Vol. 139)*. 6357-6368.

Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. 2020. Federated Optimization in Heterogeneous Networks. In *Proceedings of Machine Learning and Systems (MLSys)*, Vol. 2. 429-450.

Xiangru Lian, Ce Zhang, Huan Zhang, Cho-Jui Hsieh, Wei Zhang, and Ji Liu. 2017. Can Decentralized Algorithms Outperform Centralized Algorithms? A Case Study for Decentralized Parallel Stochastic Gradient Descent. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.

Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS) (PMLR, Vol. 54)*. 1273-1282.

H. Brendan McMahan, Daniel Ramage, Kunal Talwar, and Li Zhang. 2018. Learning Differentially Private Recurrent Language Models. In *International Conference on Learning Representations (ICLR)*.

Aditya Krishna Menon, Sadeep Jayasumana, Ankit Singh Rawat, Himanshu Jain, Andreas Veit, and Sanjiv Kumar. 2021. Long-tail Learning via Logit Adjustment. In *International Conference on Learning Representations*.

Jean Ogier du Terrail, Samy-Safwan Ayed, Edwige Cyffers, Felix Grimberg, Chaoyang He, Regis Loeb, et al. 2022. FLamby: Datasets and Benchmarks for Cross-Silo Federated Learning in Realistic Healthcare Settings. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, Vol. 35.

Teresa Salazar, Miguel Fernandes, Helder Araújo, and Pedro Henriques Abreu. 2023. FAIR-FATE: Fair Federated Learning with Momentum. In *Computational Science: ICCS 2023 (Lecture Notes in Computer Science, Vol. 14073)*. Springer, Cham, 524-538. https://doi.org/10.1007/978-3-031-35995-8_37

Canh T. Dinh, Nguyen Tran, and Josh Nguyen. 2020. Personalized Federated Learning with Moreau Envelopes. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33.

Mingxing Tan and Quoc V. Le. 2019. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. In *Proceedings of the 36th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 97)*. 6105-6114.

Stefanie Warnat-Herresthal, Hartmut Schultze, Krishnaprasad Lingadahalli Shastry, Sathyanarayanan Manamohan, Saikat Mukherjee, Vishesh Garg, et al. 2021. Swarm Learning for decentralized and confidential clinical machine learning. *Nature* 594 (2021), 265-270. https://doi.org/10.1038/s41586-021-03583-3

Frank Wilcoxon. 1945. Individual Comparisons by Ranking Methods. *Biometrics Bulletin* 1, 6 (1945), 80–83. https://doi.org/10.2307/3001968